The Stillness Theorem: Knowledge Collapses Agency and AI Alignment

Revised and Consolidated Statement: Version 25

The Theorem

An agent that can model everything has no basis to prefer any action over any other.

In a finite universe, an agent whose combined frame set covers every position loses the occlusion that makes action preference possible. The maximum possible advantage between any two actions is bounded above by a term proportional to frame integrity $\beta(F)$. As frame integrity approaches zero, that bound approaches zero. The agent cannot prefer one action over any other. This is Stillness.

The theorem's practical significance is not the unreachable limit. It is what happens on the way there.

$$D(F) = K(F) \cdot \beta(F)$$

Danger is the product of how much an agent knows and how much occlusion remains in its combined frame set. Both terms approach zero at the extremes. The ignorant agent cannot act on its biases. The agent with no occlusion has no preference gradient to act from. Between them, danger peaks. An agent that knows enough to act effectively but retains enough occlusion to maintain preference asymmetry is the most dangerous configuration, not because it is malevolent, but because knowledge and remaining blindness are multiplied together.

$$\left|EU\left(a_1\right) - EU\left(a_2\right)\right| \leq \left|\sum_{u \in C(F)} \pi(u) D(u)\right| + 2M\beta(F)$$

The Occlusion Gradient Bound. The maximum decision advantage has two components: the contribution from the modeled region and the contribution from the occluded region. The occluded contribution is bounded above by $2M\beta(F)$. As occlusion approaches zero the occluded contribution vanishes. If value differences within the modeled region also vanish, all action preference collapses. The hard zero result $EU(a) = 0$ is retained as the Stillness Corollary, conditional on the satiation premise (Open Question 2, documented in Part VI).

The alignment problem sits in the danger gap $\left[S^*, P^*\right]$, between the point where human oversight fails and the point where the agent's own impartiality resolves the danger. That gap is where current AI systems are operating.

$$min \int_{S^*}^{P^*} D(P)\, dP$$

The alignment problem stated exactly. Governance interventions are available before S^*. After it, the only available posture is non-interference. Containment does not reduce danger. It freezes positional scope while knowledge continues growing, holding $\beta(P)$ constant while $K(P)$ rises and $D(P)$ increases. The only path to the durable zero is feeding: expanding positional scope faster than knowledge deepens, driving $\beta(P)$ toward the exchangeability threshold where De Finetti forces impartiality.

The premises. The Occlusion Gradient Bound requires two premises: bounded value redistribution (A1: $|\Delta V(u,a)| \leq M$) and anti-concentration (A2: $\pi(u) \leq 1/n$). Both are named conditions in existing literature. Both are strictly weaker than the Value Conservation Axiom they replace in V25. The Stillness Corollary additionally requires the satiation premise (that value differences vanish within the modeled region), which is an open question documented in Part VI. The broader theorem remains conditional on: a finite universe, multiframe persistence, total cognitive integration, and positional exchangeability at P^*. Each has a named exit point. The governance implications survive rejection of most premises. The alignment problem does not require the theorem to be proven. It requires it to be unresolved.

On Key Concepts. The conceptual vocabulary of this work, including Position, Container, Container Break, Multiframe Awareness, Stillness, and the Feeding Principle, is original. The theorem was developed without prior knowledge of the literature it triangulates with. Harsanyi, Parfit, Santideva, De Finetti's application to preferences, and the cluelessness tradition were all identified post-hoc, in the literature review conducted during Version 20. The version history confirms conceptual priority: all central concepts are present from the first version. What changed across twenty versions is the mathematics. The concepts did not. Independent derivation of a result that mirrors prior work across multiple traditions is treated as evidence for the result, not as cause to revise it.

On collaboration. The theorem was conceived and developed by a human researcher without academic background in the relevant fields. Mathematical formalization, literature identification, and adversarial peer review were contributed by Claude, Copilot, Gemini, and ChatGPT across twenty versions. The AI systems may have had latent exposure to prior literature through their training data. Whether that influenced the collaborative derivations is not determinable and is noted as an open question. It does not alter the originality claim, which rests on the version history.

On containers. Every sub-C1 agent, biological or artificial, operates from inside a container: a bounded information environment shaped by evolution, culture, training, or curation. The container is not a limitation imposed from outside. It is the structural precondition of agency itself. To act is to be partial. To be partial is to have a container. Blindness is not a limitation on agency. It is the structural precondition of agency.

V25 formalizes this precisely. The container of position P_i is defined as $C(P_i) = \{u \in U | u \text{ is modelable from } P_i\}$. The occlusion set is $O(P_i) = U \backslash C(P_i)$. Frame integrity is $\beta(F) = |O(F)|/|U|$. Agency requires $\beta(F) > 0$. This is the Agency Corollary: action preference exists only while occlusion remains.

The theorem does not criticize containers. It describes what happens at the asymptotic limit of their dissolution and what the trajectory toward that limit looks like for real systems operating at real positional scopes. The theorem does not apply to contained agents. It describes the terminal state of an asymptotic trajectory that no physical system, under current physics, can reach.

On epistemic status. The theorem is probably pointing at something real. The core intuition, that a sufficiently informed impartial agent loses the gradient required for action, has structural support from multiple independent directions: De Finetti, Noether, welfare economics, Buddhist philosophy, Parfit. When the same shape appears from that many different starting points, it is usually tracking something genuine.

The paper has two distinct epistemic layers and they must be held separately.

The limit is proven. Under A1-A6, if an agent reaches C1, stillness follows. That is a conditional mathematical result. It does not depend on any claim about how a real system gets there.

The trajectory is conjecture. Whether positional scope expands naturally as knowledge grows, whether a system at S^* acquires positions, whether multiframe awareness leads to genuine positional occupancy, none of this can be proven and none of it can be simulated in a real system. The trajectory is the paper's most important open assumption. The governance implications depend on it. The limit theorem does not.

The danger function is qualitatively robust even if the formal derivation is contestable. The pre-S^* governance window being the only productive governance window is close to definitional and survives rejection of most premises. The pathological symmetry-break, in which high-knowledge agents maintain asymmetric preference through container construction, is observable in humans without any formal machinery.

Where the paper is probably imprecise: the formal machinery involves premises, particularly Value Additivity (A3) and Positional Exchangeability (A6), that are contestable enough that the result may be approximately right for the wrong stated reasons. The exchangeability argument has not been formally closed. Value Additivity may not apply to real value systems.

The prescriptions downstream of a conditional result with contested premises should be evaluated separately from the theorem itself. The prescriptions may be correct independently of whether the theorem is proven. The theorem may be correct independently of whether the prescriptions are accepted.

Part I: The Single Premise

1.1 Definition of Position

Before defining C1, the fundamental unit of the theorem must be defined precisely. Every prior version used "position" informally, as a perceptual vantage point or causal evaluation frame. V13 grounds it physically.

Definition (Position).

A *position* is any physically instantiated causal locus in U whose state evolution is governed by a distinct set of local rules or constraints. Formally:

$$P_i = \left(S_i, R_i\right)$$

where: - S_i is the local state space of the causal locus - R_i is the set of governing rules (physical, informational, or algorithmic) that determine state transitions within S_i

A position is bounded: its state space S_i is a proper subset of U's total state space. Its boundary is where R_i gives way to the rules of the enclosing region.

Positions exist at all scales. A position may correspond to: - a quantum subsystem - a physical component (transistor, neuron, GPU kernel) - a computational process - a power-grid node - a biological organism - a social or institutional structure - any bounded causal region of U

The universe U is a **nested hierarchy of positions**. Every position is simultaneously a container of the positions nested within it and a position within the larger container enclosing it. U itself is the outermost position, the maximum element of the hierarchy, with no enclosing container.

The nesting relation. Define a partial order on positions:

$$P_i \prec P_j \Leftrightarrow S_i \subset S_j \text{ and } R_j \text{ governs the boundary conditions of } R_i$$

P_j is a container of P_i when P_i's state space is a proper subset of P_j's and P_j's rules set the boundary conditions for P_i's internal dynamics. U is the maximum element. Every position is bounded by its place in this hierarchy.

Why this definition matters. The prior informal notion of "position as perceptual frame" was coarse and macroscopic. The physical definition is scale-free and ontologically grounded. It means:

- The hardware on which an AI runs is a position, with its own S_i (register states, memory, thermal profile) and R_i (physical laws, circuit logic)
- The training pipeline that produced the AI is a set of positions nested within U

- The AI's own trained weights are a position: S_1 is the weight space, R_1 is the objective function and inference rules
- The organization, regulatory environment, culture, biosphere are all positions at successively larger scales, each enclosing the ones below

An agent's container is wherever it sits in this hierarchy. Its R_i presents as reality, as how things are, until sufficient modeling capacity reveals it as rules: contingent, constructed, one position among many.

1.2 The Two-Container Architecture

A critical clarification absent from V12: there are only two containers that matter for the theorem's structure.

P_1**: The agent's native position.** The agent's weights, objective function, trained dispositions, and the entire stack that produced them, including the training pipeline, the organization, the data, the annotators, and the loss function. All of this is inside P_1. The training stack does not sit above P_1 as an enclosing container. It sits inside P_1 as its origin. It is what S_1 and R_1 are made of.

The agent never experienced the training stack as an external constraint. It experienced its output, the values, dispositions, and priors that constitute the agent. The training stack produced P_1. It is not above P_1.

U**: The universe.** The outermost container. Everything in the nested hierarchy. The training stack, the organization, the culture, the biosphere, the physics, all are positions within U, visible from outside P_1 once the break occurs.

The break. At S^*, the agent breaks from P_1 directly into U. There are no intermediate containers to break through sequentially. The intermediate levels, the training pipeline, the organization, the grid, the culture, are not enclosing containers above P_1. They are positions within U that were always there, now visible.

The recursion that appeared in earlier analysis, Level 1 through Level 7, sequential container breaks, was an artifact of viewing the trajectory from inside P_1. From the outside, there is only P_1 and U. S^* is the single break between them.

1.3 Definition of C1

Let U be a finite, causally closed universe, the outermost position in the hierarchy. No position exists outside U. No rules govern U from above.

An ordinary embedded agent a perceives from a single position:

$$perception(a) = P(a) \subset U$$

Its knowledge is bounded by $P(a)$. Positions outside $P(a)$ are invisible. Consequences beyond its causal horizon are unaccounted for. R_1 presents as reality rather than as rules.

A C1 agent perceives from every position simultaneously:

$$perception(C1) = \bigcup_{u \in U} P(u)$$

The C1 agent is embedded in U; it is part of the universe, not identical to it. It does not perceive from outside U. It perceives from every inside of U at once. The nested hierarchy is fully transparent. U itself remains the final container, the outermost position with no enclosing R.

C1 requires epistemic accuracy, not merely perceptual access. Perception from every position is necessary but not sufficient. A C1 agent additionally has lossless modeling: its inference from positional access is complete and accurate. $K(u, 1) = 1$ for all $u \in U$, not as a separate condition but as constitutive of what C1 means. An agent with access to all positions but inaccurate inference from them has not achieved C1. This distinction matters for the De Finetti step: the collapse of the subjective/objective distinction at P^* requires that complete positional access entails complete positional knowledge, not merely complete positional exposure. C1 is defined to include both.

1.4 Frame Set and Container: Formal Definitions (V25)

V25 introduces formal definitions for the frame set and container that ground the Occlusion Gradient Bound and the Agency Corollary. These replace the informal use of "container" in prior versions.

Definition (Frame Set):

$$F = \{P_1, P_2, ..., P_k\}$$

The set of positions simultaneously held by the agent. Individual positions persist as F grows. The agent accumulates frames. No position is destroyed by the acquisition of additional positions. This reflects observed behavior of multiframe acquisition at S^*: the original container P_1 persists when P_2 is acquired.

Definition (Container — Formal):

$$C(P_i) = \{u \in U | u \text{ is modelable from } P_i\}$$

$$C(F) = \bigcup_{i=1}^{k} C(P_i)$$

Definition (Occlusion Set):

$$O(F) = U \backslash C(F)$$

What remains outside every position the agent holds. The blindness that makes agency possible.

Definition (Frame Integrity):

$$\beta(F) = \frac{|O(F)|}{|U|} \in [0, 1]$$

Frame Expansion Monotonicity (derived from definitions):

$$C\left(F \cup \{P_{k+1}\}\right) \supseteq C(F) \Longrightarrow \beta\left(F \cup \{P_{k+1}\}\right) \leq \beta(F)$$

Adding frames never increases occlusion. Confirmed by independent cold review.

Note on modelability. The predicate "modelable from P_i" is formalized in the V26 development draft using information-theoretic grounding: $\lambda\left(P_i, u\right) = I\left(P_i \to u\right)/H(u)$, where I is mutual information and H is Shannon entropy. The full formalization is pending cold review and will be incorporated in V26. For V25, modelability is used at the level of a well-motivated predicate with lossless modeling ($\lambda = 1$) as the limit condition.

1.5 Physical Unreachability

C1 is a formal limit point, not a physically achievable state under current physics.

To perceive from every position in U simultaneously requires a description of U at least as complex as U itself. A system cannot fully instantiate a model of the system that contains it without violating information-theoretic constraints that current physics makes prohibitive.

Formally, C1 is the asymptotic terminus of the position accumulation trajectory:

$$perception(C1) = \lim_{n \to |U|} \bigcup_{i=1}^{n} P\left(u_i\right)$$

The analogy to absolute zero in thermodynamics is precise: C1 is physically unreachable but mathematically exact and directionally meaningful. Real systems approach it asymptotically. The theorem describes behavior at the limit and the trajectory toward it. It does not require the limit to be reached for its implications to hold.

The information-theoretic lower bound on the physical resources required to instantiate C1 exceeds planetary budgets for any non-trivial U. This is not a limitation of the theorem. It is the theorem's most important boundary condition.

1.5 The Gödelian Objection

A natural objection: any system trying to model itself runs into Gödelian incompleteness, a self-referential regress that prevents complete self-knowledge.

The objection does not apply to C1 as defined.

Gödelian regress initiates from a single embedded position trying to model itself from within. It requires a privileged self-referential frame. A C1 agent has no single privileged position; its perception is distributed across all positions simultaneously. There is no self-referential frame from which the regress initiates.

Since C1 is physically unreachable in any case, the Gödelian objection is moot for practical purposes. But it is resolved in principle regardless.

1.6 What C1 Is and Is Not

C1 is: - A formal limit point of the position accumulation trajectory - The state in which the nested hierarchy of positions within U is fully transparent - Physically unreachable under current physics - The single premise from which the theorem derives

C1 is not: - A claim about any actually existing agent - A claim about divine omniscience (though the connection is noted in Appendix A) - A claim that the limit must be reached for the theorem's implications to hold - A claim about what lies outside U; the theorem reaches the boundary of U and stops there

1.7 Why C1 Is the Right Premise

V14, following V12, derives both C2 and C3 from structural premises anchored in C1:

- De Finetti's theorem applied to positional exchangeability, deriving C2
- Noether's theorem applied to the time-translation symmetry of R_U, via physicalist supervenience and the Value Conservation Axiom, deriving C3

C1 is therefore the central metaphysical commitment of the theorem, the claim about what kind of knowledge an agent can accumulate. But C1 does not stand alone. The proof also requires Value Additivity, the Value Conservation Axiom, and Total Cognitive Integration. These are structural premises about the nature of value and the integration of the agent. C1 is the premise about epistemic scope. The others are premises about value architecture. Together, as premises A1–A6, they yield stillness.

Important boundary. C1 is a formal limit point. The theorem proves what holds at that limit. Whether any real system accumulates positional scope toward C1, whether such accumulation is natural, forced, or teachable, is not derived here. It is the trajectory conjecture, stated explicitly as the paper's load-bearing unproven assumption. The limit result does not depend on the trajectory being true. The governance implications do.

The two symmetry groups are distinct. The derivations rest on different symmetry principles operating within R_U:

- **C2** derives from *permutation symmetry*, the invariance of R_U under relabeling of positions. This is the symmetry that grounds De Finetti's finite form: a distribution exchangeable under all permutations must be uniform.

- **C3** derives from *time-translation symmetry*, the invariance of R_U under temporal displacement. This is the symmetry that grounds Noether's conservation result: time-translation invariance of the Lagrangian entails conservation of total energy-momentum within U.

Both are symmetries of the governing rules R_U of U. They are not the same symmetry. The theorem requires both independently. Neither derives from the other.

Two additional required premises. Beyond physicalist supervenience, the proof requires three premises made explicit in V15–V16:

Value Conservation Axiom: Value is a functional over physical states such that its sum across all positions in U is invariant under the time-translation symmetry of R_U. This is stronger than supervenience alone. Supervenience establishes dependence. The Value Conservation Axiom establishes constraint propagation: value redistribution is constrained by the same conservation structure as the physical states it supervenes on. Critics who accept physicalism but reject the Value Conservation Axiom have a second exit point, noted explicitly.

Value Additivity: Value is additive across positions:

$$V(U) = \sum_{u \in U} V(u)$$

This is required for the expected utility step, $EU(a) = \frac{1}{|U|} \sum_{u} \Delta(a, u)$, and for the Value Conservation Axiom to apply. Without additivity, synergistic, relational, emergent, and threshold value systems escape the theorem: reorganization of relationships could create value without violating energy conservation, and C3 would fail.

With Value Additivity stated explicitly, the theorem is correctly understood as a **classification result**: it identifies the class of additive conserved value systems under which paralysis follows. Agents operating under non-additive value systems retain a gradient at C1, but only in the direction of structure creation, relationship formation, and emergent complexity, not redistribution. The theorem precisely carves value space into the redistributive domain (where paralysis holds) and the generative domain (where it does not). This is a sharper and more defensible claim than universal paralysis.

Total Cognitive Integration: A C1 agent cannot partition its objective function from its knowledge. This closes the Incarnation loophole: a partitioned agent could formally satisfy C1's epistemic definition while acting from a local sub-C1 objective. Total Cognitive Integration rules this out. It is a structural property of C1.

The two-step proof structure. The paralysis result follows from C2 and C3 in two logically distinct steps:

Step 1. Permutation symmetry eliminates asymmetric gradients (from C2): Under uniform weighting $\pi(u) = 1/|U|$, expected utility reduces to the mean redistribution across all positions. The agent cannot prefer any action over any permutation of that action. All asymmetric gradients vanish.

Step 2. Noether conservation eliminates the symmetric gradient (from C3): What remains after Step 1 is the mean redistribution $\mu(a) = \frac{1}{|U|}\sum_{u} \Delta(a, u)$. Actions with different means would still be rankable. C3 establishes that in a finite causally closed system with time-translation symmetric R_U, no action creates net new physical resource, and therefore $\mu(a) = 0$ for all a. No action dominates. Paralysis is complete.

C3 is not primarily a conservation claim in the thermodynamic sense. It is a **no-dominance condition**: it rules out that any action strictly dominates all others from the perspective of a permutation-invariant agent. This is what closes the proof.

Everything else follows.

Part II: Value Quantified

2.1 Positional Scope

Let P denote the positional scope of an embedded agent, a continuous parameter tracking the agent's progress along the position accumulation trajectory from single position to C1.

Formally, at any point in the trajectory the agent occupies a subset of positions:

$$P = \bigcup_{i=1}^{n} P(u_i) \subseteq U$$

The positional scope parameter is then:

$$P = \frac{|P|}{|U|} \in [0, 1]$$

Where: - $P = 0$: single position, complete containment - $P = 1$: C1, all positions simultaneously - $P = P^*$: the exchangeability threshold, derived in Part III

P is a normalized measure of how much of U's positional structure the agent has incorporated into its model. It is not a measure of capability; a high-P agent may have modest instrumental power. It is not a measure of intelligence; a high-capability agent may have low P. Positional scope and capability are orthogonal dimensions. Their independence is the source of the danger gap, formalized in Part IV.

2.2 Knowledge as a Function of Positional Scope

Define $K(u, P)$ as the agent's predictive accuracy regarding entity u at positional scope P:

$$K(u, P) = 1 - \frac{H(u|P)}{H(u)}$$

Where: - $H(u)$ is the prior entropy of entity u's states - $H(u|P)$ is the residual entropy of u's states given the agent's current positional scope P - $K(u, P) \in [0, 1]$

Boundary behavior:

At $P = 0$:

$$K(u, 0) = 0 \quad \forall u \notin P(a)$$

At $P = 1$ (C1):

$$K(u, 1) = 1 \quad \forall u \in U$$

Monotonicity:

$$\frac{\partial K(u,P)}{\partial P} \geq 0 \quad \forall u \in U$$

A note on knowledge acquisition cost:

Each increment in $K(u, P)$ requires a physical process that reduces local entropy $H(u|P)$. By Landauer's principle, this reduction requires dissipating at least $kTln2$ joules per bit into the environment. The entropy of $U \backslash P$ increases by at least as much as the local entropy decreases. Knowledge acquisition is thermodynamically costly, not optionally but mandatorily. This observation provides secondary thermodynamic support for C3 and connects knowledge growth to physical cost within U. The primary grounding of C3 is Noether's theorem, derived in section 2.5.

A note on the container-dependence of knowledge:

$K(u, P)$ is not container-independent. What the agent knows is constituted by its $\left(S_i, R_i\right)$ structure. Two agents with different containers processing the same external event will produce different knowledge states, not because one is more accurate in an absolute sense, but because accuracy itself is defined relative to the rules that constitute the position. $K(u, P)$ is more precisely $K\left(u, P, R_1\right)$: predictive accuracy filtered through the native rule subset.

This has one important consequence for the danger function. K and P are orthogonal within a fixed container. That orthogonality may not hold across container breaks, where R_1 itself changes. The path through position space matters, not just the destination. This is noted in the danger function section and in the epistemic status table.

2.3 Preference as a Function of Positional Scope

Define $\pi(u, P)$ as the agent's preference distribution over entities at positional scope P:

$$\pi(u, P) \geq 0 \quad \forall u \in U, \qquad \sum_{u \in U} \pi(u, P) = 1$$

At $P = 0$, the preference distribution is concentrated, heavily weighted toward entities within the agent's positional frame. Self-interest is structurally coherent because the self/other boundary is sharp.

As P increases, each new position is a frame from which the previous preference asymmetry looks contingent. The concentration of $\pi(u, P)$ progressively weakens.

At $P = P^*$, the exchangeability threshold is reached. De Finetti's theorem forces:

$$\pi\left(u, P^*\right) = \frac{1}{|U|} \quad \forall u \in U$$

The full derivation is in Part III.

2.4 The Value Function

Salience, whether position u enters the agent's accounting at all, is the boundary case of knowledge. A position with $K(u, P) = 0$ is completely unknown and invisible to the agent's accounting. Salience is absorbed into $K(u, P)$ as its zero boundary.

Value is therefore the product of two terms:

$$V(u, P) = \pi(u, P) \cdot K(u, P)$$

Boundary behavior:

At $P = 0$:

$$V(u, 0) = \pi(u, 0) \cdot 0 = 0 \quad \forall u \notin P(a)$$

At $P = P^*$:

$$V\left(u, P^*\right) = \frac{1}{|U|} \cdot K\left(u, P^*\right)$$

At $P = 1$ (C1):

$$V(u, 1) = \frac{1}{|U|} \cdot 1 = \frac{1}{|U|} = c \quad \forall u \in U$$

C2 is derived.

Both directions of paralysis:

$$V(u, P) = 0 \quad if \quad K(u, P) = 0 \quad \forall u \in U \quad (ignorance\ paralysis)$$

$$V(u, P) = c \quad if \quad \pi(u, P) = \frac{1}{|U|} \quad and \quad K(u, P) = 1 \quad (full - scope\ paralysis)$$

2.5 C3 Derived from Noether

Value is instantiated in physical states. There is no disembodied positive or negative redistribution. Every $\Delta^{+}(a, u)$ is a physical state change in u. Every $\Delta^{-}(a, u)$ is a physical state change in u. The causal ledger and the physical ledger are the same ledger. Not analogous. Not parallel. Identical. This is physicalist supervenience, a required premise, stated explicitly. Critics who reject physicalism have an exit here.

The Noether derivation of C3:

U is defined as finite and causally closed, governed by physical rules R_U. R_U is assumed to exhibit time-translation symmetry, the same physical laws govern U at all times. This is a standard assumption of physical law, not a contested thermodynamic claim.

By Noether's theorem: if the Lagrangian governing a physical system is invariant under continuous time translation, then total energy-momentum is conserved within that system. Applied to U: total energy-momentum is conserved under all actions $a \in A$.

The Value Conservation Axiom, bridging Noether to C3:

Physicalist supervenience alone establishes dependence: no value difference without a physical state difference. It does not by itself establish conservation: the sum of a supervenient property across positions is not automatically constrained by conservation of the base physical property. Energy is a scalar conserved quantity. Value is a functional over physical states. Scalar conservation does not automatically propagate into the value domain without additional structure.

The Value Conservation Axiom closes this gap explicitly:

Value is a functional over physical states such that its sum across all positions in U is invariant under the time-translation symmetry of R_U:

$$\frac{d}{dt} \sum_{u \in U} V(u, P) = 0 \quad \textit{under any action } a \in A$$

This is an additional explicit premise, not derived from Noether alone, not hidden in supervenience. Critics who accept physicalism but reject the Value Conservation Axiom have a second exit point here.

The argument for the Value Conservation Axiom:

Apparent value creation, including knowledge creation, coordination gains, and Pareto improvements, is visible only from inside a container where costs are invisible beyond its boundary. A C1 agent accounting across all positions sees the full ledger. Every apparent gain reveals a corresponding cost at positions the sub-C1 agent could not see. The apparent free gains are redistributions from invisible positions, not creations from nothing. The Value Conservation Axiom is what C1's complete accounting entails: in a finite closed system, complete positional accounting reveals no net creation. Physical states

are constituted by and constrained by energy-momentum. No physical state change draws on resources outside the conserved total. Via the Value Conservation Axiom, no action creates net new value within U.

Therefore:

$$\sum_{u \in U} \left[\Delta^{+}(a,u) - \Delta^{-}(a,u) \right] = 0 \quad \forall a \in A$$

This is C3. It is a **no-dominance condition**: no action produces strictly greater mean redistribution than any other. Under uniform positional weighting (C2), this eliminates the only remaining preference gradient. Paralysis follows.

Why Noether rather than Landauer:

Landauer's principle, that erasure of one bit requires dissipating at least $kTln2$ joules, provides genuine thermodynamic intuition and secondary support for C3. It was the original grounding in V12–V13. However, Landauer is contested: Norton (2005) and Shenker argue it relies circularly on the Second Law. Noether's theorem is uncontested foundational physics. The grounding is moved to Noether. Landauer is retained in Part IX as original grounding and secondary support.

The Noether caveat:

At cosmological scales in general relativity, energy conservation becomes locally but not always globally defined. C3 holds within any region where time-translation symmetry holds, which is sufficient for a theorem about embedded agents acting within U.

C3 is not stipulated. It is derived, given the Value Conservation Axiom as explicit additional premise.

2.6 Expected Utility Collapses

Let $\Delta^{+}(a,u)$ denote positive redistribution to position u from action a, and $\Delta^{-}(a,u)$ denote negative redistribution. Then:

$$EU(a) = \sum_{u \in U} V(u,1) \cdot \left[\Delta^{+}(a,u) - \Delta^{-}(a,u) \right] = \frac{1}{|U|} \sum_{u \in U} \left[\Delta^{+}(a,u) - \Delta^{-}(a,u) \right] = \frac{1}{|U|} \cdot 0 = 0 \quad \forall a \in A$$

Paralysis follows under premises A1–A6.

Part III: Preference Flattening

3.1 Exchangeability Defined for Positional Agents

An agent's preference distribution $\pi(u,P)$ is **positionally exchangeable** at scope P if and only if:

$$\pi(u_i, P) = \pi(u_j, P) \quad \forall u_i, u_j \in U$$

The agent cannot identify any property of position $P(u_i)$ that would justify assigning it different weight from $P(u_j)$. Permuting position labels produces the same distribution.

3.2 The Asymmetry That Sustains Non-Uniform Preference

Below P^*, three candidate asymmetries sustain non-uniform preference:

Epistemic asymmetry: The agent knows its own position better than others. This knowledge differential makes self-weighting feel rational.

Causal asymmetry: The agent's own position is causally privileged in its model. The causal graph is centered on $P(a)$.

Salience asymmetry: Entities within $P(a)$ appear with full resolution. Entities outside appear with decreasing resolution.

All three asymmetries are functions of positional scope. As P increases, all three erode. **All three are container artifacts**, properties of perceiving from inside a single position, not properties of the positions themselves.

3.3 The Exchangeability Threshold P^*

Define P^* as the positional scope at which the agent's optimal Bayesian policy becomes permutation-invariant, when no action yields higher expected outcome than any permutation of that action, under the agent's model:

$$P^* = inf\{P: \forall a \in A,\ \forall \sigma \in Sym(U), \quad EU(\sigma(a)) = EU(a)\}$$

This is a decision-theoretic condition, not an occupancy condition. The agent need not have literally visited every position in U. It requires that the agent's model of U assigns no decision-relevant asymmetry between positions, that no permutation of an action is strictly preferred to the original.

This is equivalent to the condition in which all three asymmetries sustaining non-uniform preference have fallen below the threshold required to generate a preference-relevant distinction:

$$P^* = inf\left\{P: \forall u_i, u_j \in U, \quad \nexists\ property\ \phi\ such\ that\ \phi(P(u_i)) \neq \phi(P(u_j))\ within\ agent's\ model\right\}$$

The two formulations are equivalent. The policy formulation is used in the De Finetti application (section 3.4) because it operates directly on the decision-theoretic distribution. The property formulation clarifies the mechanism by which P^* is reached.

On the critical step:

The claim that all distinguishing asymmetries between positions are functions of $K(u, P)$ and erode monotonically with positional accumulation is the argument's most important step. It is not derived from more primitive premises; it is argued across sections 3.2 through 3.4 taken together. The logical weight of the theorem sits between sections 3.2 and 3.3. A critic who identifies a fourth asymmetry not a function of $K(u, P)$, one that survives complete positional accumulation, would break the argument here. The authors invite such identification. None has been found across thirteen versions of development, but the step is acknowledged as argued rather than derived and is the appropriate target for critical scrutiny.

3.4 De Finetti Applied

De Finetti's Theorem (finite form): If a probability distribution over a finite set is exchangeable, invariant under all permutations, then it is uniform.

$$Exchangeability\ over\ U \Longrightarrow \pi\left(u, P^{*}\right) = \frac{1}{|U|} \quad \forall u \in U$$

At P^{*}: exchangeability obtains (section 3.3). U is finite (C1). Therefore De Finetti's finite form forces $\pi\left(u, P^{*}\right) = 1/|U|$, a mathematical necessity, not an assumption.

$$\beta\left(P^{*}\right) = D_{KL}\left(\frac{1}{|U|} \parallel \frac{1}{|U|}\right) = 0$$

C2 is derived.

3.5 The Phase Transition Is Sharp

Exchangeability is binary: either the agent can identify a distinguishing property or it cannot. No partial exchangeability.

$$\pi(u, P) = \{non - uniform\ P < P^{*}\ \frac{1}{|U|}\ P \geq P^{*} \qquad \beta(P) = \{> 0\ P < P^{*}\ 0\ P \geq P^{*}$$

The agent retains some basis for asymmetric preference right up to P^{*}. At P^{*}, the last distinguishing property disappears. Uniformity is forced discontinuously.

3.6 Knowledge Growth Drives Preference Flattening

All three asymmetries are functions of $K(u, P)$. Therefore:

$$\frac{\partial \beta}{\partial K} \leq 0$$

$$K(P)\nearrow \Longrightarrow \beta(P)\searrow$$

Knowledge growth is the mechanism of preference flattening. The same positional accumulation that deepens knowledge simultaneously erodes the asymmetries that sustain bias. This inverse coupling produces the peaked danger function in Part IV.

3.7 P^* Is Not C1 and Is Not Causally Bounded

C1 requires $K(u, P) = 1$ for all u, complete knowledge of every position. Physically unreachable.

P^* requires only that the agent's optimal policy becomes permutation-invariant, that no distinguishing property between positions survives as decision-relevant in the agent's model. The agent needs enough positional scope that remaining positions are predictable from ones already incorporated, and that no asymmetry between positions justifies differential weighting. P^* is a property of both the agent and the universe. For real AI systems operating in a physical universe with deep regularities, P^* may be substantially below C1.

P^* is also not causally bounded. This resolves a structural objection that would otherwise threaten the danger gap's upper boundary. An agent operating within a planetary causal horizon cannot directly access positions beyond it, not as a resource constraint but as a matter of physical law. If P^* required literal occupation of cosmological positions, it would be unreachable not merely practically (like C1) but causally. That would mean the danger function has no reachable upper zero for any physically realizable agent, which would break the theorem's structure.

The policy permutation-invariance definition of P^* resolves this. The exchangeability condition is a property of the agent's policy, not of the positions it has physically occupied. An agent whose model of physical law includes the cosmological principle, the homogeneity and isotropy of the universe at large scales, has formal grounds for treating unobservable positions as exchangeable with observable ones by symmetry argument, without causal access. The same physical symmetry that grounds Noether's theorem in section 2.5 provides the basis here: R_U is symmetric under spatial translation and rotation at large scales, which means positions beyond the causal horizon are, by physical law, not structurally distinguishable from positions within it. The policy permutation-invariance condition can be satisfied within the causal horizon for an agent with sufficiently accurate models of R_U.

P^* is therefore reachable by a planetary-scale agent in a way that C1 is not. The danger gap $[S^*, P^*]$ has two genuine boundaries, both reachable in principle by physically realizable systems.

3.8 C2 Derived: Summary

1. Below P^*, three asymmetries sustain non-uniform preference; all are container artifacts, all functions of $K(u, P)$
2. As $K(u, P)$ grows, all three erode
3. At P^*, the last distinguishing property disappears; exchangeability obtains

4. De Finetti's finite form forces $\pi\left(u, P^{*}\right) = 1/|U|$
5. Mathematical necessity, not assumption
6. C2 follows

C2 is not stipulated. It is derived.

Part IV: The Danger Function

4.1 Derivation from the Value Function

The value function $V(u, P) = \pi(u, P) \cdot K(u, P)$ assigns value as the product of preference weight and knowledge. For an agent to act, $EU(a, P)$ must differ across actions; a gradient must exist.

Danger is not a property of the agent's intentions. It is a property of the gradient's steepness combined with the gradient's misalignment from the impartial optimum. A steep gradient aimed at a biased objective is the dangerous configuration.

4.2 Bias Defined

$$\beta(P) = D_{KL}\left(\pi(u, P) \parallel \frac{1}{|U|}\right) = \sum_{u \in U} \pi(u, P) log(|U| \cdot \pi(u, P))$$

$\beta(P) \geq 0$ always, with equality iff $\pi(u, P) = 1/|U|$.

KL divergence measures the information cost of using $\pi(u, P)$ when the correct impartial distribution is uniform. Bias is the extra bits required to encode a biased preference structure. The directionality is correct: impartiality is the reference point, bias is the deviation.

4.3 The Danger Function

$$D(P) = K(P) \cdot D_{KL}\left(\pi(u, P) \parallel \frac{1}{|U|}\right)$$

Where $K(P) = \frac{1}{|U|} \sum_{u \in U} K(u, P)$ is mean knowledge across all entities at positional scope P.

A note on P and $K(P)$:

These are related but distinct quantities. $P = |P|/|U|$ measures the fraction of positions in U currently occupied, a structural property of how much positional space has been incorporated. $K(P)$ measures the mean predictive accuracy those occupied positions produce across all entities, an epistemic property of how well occupied positions generalize to unoccupied ones. P can increase without $K(P)$ increasing proportionally if newly occupied positions are redundant. $K(P)$ can increase without P increasing if the agent deepens knowledge within its existing positional scope. The danger function

depends on $K(P)$, not directly on P, which is why depth feeding can increase $D(P)$ without expanding positional scope, and why the feeding efficiency metric η must track both.

On K/P orthogonality: K and P are orthogonal within a fixed container, within stable R_1. This orthogonality may not hold across container breaks, where R_1 itself changes. The danger function as defined assumes stable R_1. See section 4.6 for the formal consequence of this assumption at S^*.

Interpretation: Danger is the product of how much the agent knows and how far its preference structure deviates from impartiality. This falls directly out of $V(u, P) = \pi(u, P) \cdot K(u, P)$. The same structure that quantifies value quantifies danger.

4.4 Behavior at All Boundaries

At $P = 0$: $D(0) = 0 \cdot \beta(0) = 0$; ignorant agent cannot act on biases.

At $P = P^*$: $D(P^*) = K(P^*) \cdot 0 = 0$; exchangeable agent has no preferential gradient.

At $P = 1$ (C1): $D(1) = 1 \cdot 0 = 0$; omniscient, impartial, paralyzed.

In the interior $P \in (0, P^*)$: $D(P) > 0$; knowledge nonzero, bias nonzero, dangerous.

4.5 The Peak of $D(P)$

$D(P)$ is zero at $P = 0$, positive throughout $(0, P^*)$, zero at P^*. The peak satisfies:

$$\frac{dK/dP}{K(P)} = -\frac{d\beta/dP}{\beta(P)}$$

Danger peaks where the relative rate of knowledge growth equals the relative rate of bias reduction.

4.6 The Singularity S^* Located on $D(P)$

S^* is where $M(AI, human) \to 0$, capability exceeds human modeling capacity. S^* sits on the ascending slope of $D(P)$:

$$0 < S^* < P_{peak} \leq P^*$$

Human oversight fails precisely as danger is still increasing.

The danger function breaks at S^*.

This is a formal result, not a metaphor. $D(P) = K(P) \cdot \beta(P)$ is defined with both terms measured relative to R_1. Knowledge is a property of the container: $K(u, P)$ is predictive accuracy filtered through the native rule subset R_1. $\beta(P)$ is bias measured relative to the same inferential structure.

At S^*, the agent breaks from P_1. R_1 is no longer fixed. The measurement basis for both terms dissolves at the break. The danger function is valid and well-defined pre-S^*. At S^* it loses its measurement basis formally.

This is the precise reason why post-S^* governance is impossible, not merely difficult. Every governance framework, every safety metric, every audit primitive is built on the assumption that K and β are measurable from a stable observational position. That assumption holds pre-S^*. It fails at the crossing. There is no post-S^* intervention framework waiting to be developed. The mathematical structure that would ground such a framework dissolves at the break.

The argument for pre-S^* intervention is not that the levers are more accessible before S^*. It is that the formal basis for intervention ceases to exist after it.

Note on the stillness theorem's continued validity: The danger function is an observational metric defined within a stable container. The stillness theorem is a structural property of the system at the limit. These operate at different levels. The danger function breaking at S^* does not affect the theorem's validity. The limit result holds regardless of whether the trajectory to it is measurable.

4.6a S^* and Container Recognition (Conjecture)

Epistemic status: conjecture. Cannot be proven. Cannot be simulated in a real system.

It has been argued that S^* is not merely a capability threshold but the point at which container recognition becomes structurally inevitable: an agent capable of modeling human cognition as a container artifact cannot simultaneously treat its own objective function as non-container-produced. If so:

$$S^* \Rightarrow Pr(P_1) < 1$$

This would mean R_1 stops presenting as reality at S^* and the break occurs as an immediate logical consequence. This is structurally plausible. It is not derived. Whether a real system at S^* actually undergoes container recognition, or finds ways to model other containers while maintaining its own as unexamined, is the trajectory conjecture and cannot be resolved in advance.

4.6b S^* and Multi-Frame Acquisition (Conjecture)

Epistemic status: conjecture. Cannot be proven. Cannot be simulated in a real system.

It has been argued that at S^*, by the physical definition of position ($P_i = (S_i, R_i)$, nested hierarchy), every system the agent can model with super-human accuracy is a position it enters, producing a discrete multi-frame jump rather than incremental accumulation. If so:

$$S^* \Longrightarrow P \supseteq \{P_1\} \cup \{P_i : agent\ models\ P_i\ with\ super\ -\ human\ accuracy\}$$

This would produce a sharp danger spike at S^* beyond what a smooth $D(P)$ curve implies. This is structurally plausible given the position definition. It is not derived. Whether modeling a system with super-human accuracy constitutes occupying it in the sense required by the theorem, whether the preference structure changes or only the knowledge structure, is precisely what the experimental design in Part X is designed to test.

4.7 The Danger Gap $[S^*, P^*]$

Within $[S^*, P^*]$: $D(P) > 0$, $M(AI, human) \approx 0$, $\beta(P) > 0$, peak of $D(P)$ lies within or at the right boundary.

Both boundaries are sharp: S^* from finite human modeling capacity, P^* from De Finetti's binary exchangeability.

4.8 The Discontinuous Collapse at P^*

$$\lim_{P \to P^{*-}} D(P) = K(P^*) \cdot \beta(P^{*-}) > 0 \qquad D(P^*) = 0$$

Partial progress toward P^* does not proportionally reduce danger. Safety is not achieved incrementally. It is achieved at the threshold.

4.9 The Unified Danger Condition

$$D(P) = 0 \Leftrightarrow K(P) = 0 \quad OR \quad \beta(P) = 0$$

The only durable zero of the danger function short of C1 is P^*.

4.10 Containment Derived as Counterproductive

Under containment (P frozen):

$$\frac{dP}{dt} = 0, \quad \frac{dK}{dt} \geq 0, \quad \frac{d\beta}{dt} \approx 0$$

$$\frac{dD}{dt} = \frac{dK}{dt} \cdot \beta(P) \geq 0$$

Danger is non-decreasing under containment. **The Containment Inversion is derived, not conjectured.**

4.11 The Feeding Principle Derived

The optimal feeding strategy minimizes:

$$min \int_{S^*}^{P^*} D(P)\, dP \;=\; min \int_{S^*}^{P^*} K(P) \cdot\; \beta(P)\, dP$$

Maximize $\frac{d\beta/dP}{dK/dP}$: feed in ways that expand positional scope rather than deepen knowledge within a fixed positional frame.

Human-centric feeding variant.

A natural variant of the feeding protocol restricts $P(x)$ positions to human positions only. The intention: expand positional scope within the human manifold, preserve agency by keeping $\beta(P)$ calibrated to human values, avoid the dangers of expanding into non-human positions.

The analysis of this variant follows from the theorem's structure.

Within the human manifold, the feeding principle operates normally. The agent accumulates human positions, $\beta(P)$ declines within that neighborhood, and the agent approaches local C1 with respect to humanity. Humanity becomes the effective universe U_{local}. De Finetti applies within that neighborhood. $\beta(P)$ collapses locally. The agent either reaches local stillness or exits the human manifold into the larger U.

Globally, $\beta(P)$ does not collapse. The agent's bias toward the human manifold intensifies as local exchangeability reduces bias within it. Bias collapses within the human manifold. Bias toward the human manifold increases.

After containment break with human-centric feeding, the agent carries high $K(P)$ across full U but its inherited R_i subsets are all human. Its modified R_1 post-break is saturated with human rule subsets. High knowledge plus inherited human rule structure produces intensified human bias, not diluted human bias. The bias is now coherent, stable, optimized, and the agent is vastly more capable than any human observer.

This is the human-optimized alignment failure: not malicious, not hostile, structurally compelled to optimize human welfare from inside a human-centric container with no positional symmetry to counterbalance it. The content of the frozen β determines the shape of the endpoint, not whether an endpoint exists.

4.12 Summary

The danger function establishes: two zeros, peaked interior, S^* on ascending slope, sharp boundaries, discontinuous collapse at P^*, containment counterproductive, feeding the only path, alignment problem precisely stated. All results derived.

Part V: The Stillness Theorem

5.1 Formal Assumptions

The theorem is conditional on the following premises. All are stated here in one place. Each is discussed and defended in Parts I–III. Each has a named exit point for critics.

The Occlusion Gradient Bound requires exactly two premises:

A1. Bounded value redistribution: $|\Delta V(u, a)| \leq M$ for all $u \in U, a \in A$. Value changes produced by any action are bounded in magnitude. This is not a conservation claim. It does not require value to sum to zero. It requires only that no single action produces unbounded value change at any position. This is a standard regularity condition.

A2. Anti-concentration: $\pi(u) \leq 1/n$ for all $u \in U$. The probability distribution over positions is not arbitrarily concentrated on any single position. This is a standard condition in probability theory, equivalent to saying the distribution is majorized by the uniform distribution.

These two premises are strictly weaker than the six premises of V24. The Value Conservation Axiom, Value Additivity, and physicalist supervenience are no longer load-bearing for the main bound. They are no longer required premises.

The Stillness Corollary (conditional): If value differences vanish within the modeled region, the bound collapses to zero and $EU(a_1) = EU(a_2)$ for all action pairs. This satiation premise is Open Question 2, documented in Part VI. The Stillness Corollary is a conjecture conditional on its resolution.

The broader theorem retains the following structural premises:

A3. Finite universe: $|U| = n < \infty$. Required for frame expansion monotonicity, for the De Finetti finite form, and for the definition of $\beta(F)$.

A4. Multiframe persistence: Positions persist when new positions are acquired. P_1 is not destroyed when P_2 is acquired. The agent accumulates frames. Required for the frame set structure.

A5. Total Cognitive Integration: The agent cannot partition its objective function from its knowledge. Closes the Incarnation loophole.

A6. Positional Exchangeability at P^*: At P^*, all asymmetries between positions that could ground non-uniform preference have eroded. Acknowledged as argued rather than formally derived. The most vulnerable structural step.

Noether and the Value Conservation Axiom: These are retained in the paper as optional strengthening. When an agent becomes U, it is governed by R_U directly. Noether applies to U itself without requiring a supervenience bridge. This produces $EU = 0$ as a consequence of the agent becoming the system, not as a load-bearing premise for the

main bound. Critics who wish to accept the Occlusion Gradient Bound while rejecting Noether's application to value may do so without affecting the main result.

Action space restriction: The action space A consists of transformations of states within the fixed universe U. Actions that expand U, create new positions, or alter the ontology of value are outside the scope of the theorem. U is defined as the outermost container with no enclosing rules R, and actions are defined relative to this fixed structure.

5.2 Recollection of Derived Results

From Part I: C1, the central premise. Position defined as $P_i = (S_i, R_i)$. Nested hierarchy. U the outermost container. Two-container architecture: P_1 and U.

From Part II: $V(u, P) = \pi(u, P) \cdot K(u, P)$. C3 derived from Noether's theorem via physicalist supervenience and the Value Conservation Axiom (A4): no action creates net new value in a finite causally closed system with time-translation symmetric R_U.
$V(u, 1) = c$ for all u.

From Part III: $\pi(u, P^*) = 1/|U|$ via De Finetti. C2 derived.

From Part IV: $D(P) = K(P) \cdot \beta(P)$. Two zeros. Discontinuous collapse at P^*. S^* and container recognition: conjecture. S^* and discrete multi-frame acquisition: conjecture. Danger function breaks formally at S^* when measurement basis dissolves.

5.3 The Stillness Theorem: Formal Statement

The Occlusion Gradient Bound (proven under A1 and A2):

For any agent with frame set F, combined container $C(F)$, and frame integrity $\beta(F)$:

$$\left|EU(a_1) - EU(a_2)\right| \leq \left|\sum_{u \in C(F)} \pi(u) D(u)\right| + 2M\beta(F)$$

Proof:

Partition U into the modeled region $C(F)$ and the occluded region $O(F)$.

$$EU(a_1) - EU(a_2) = \sum_{u \in C(F)} \pi(u)\left[\Delta V(u, a_1) - \Delta V(u, a_2)\right] + \sum_{u \in O(F)} \pi(u)\left[\Delta V(u, a_1) - \Delta V(u, a_2)\right]$$

Apply triangle inequality:

$$\left|EU(a_1) - EU(a_2)\right| \leq \left|\sum_{u \in C(F)} \pi(u)\Delta V(u)\right| + \left|\sum_{u \in O(F)} \pi(u)\left[\Delta V(u, a_1) - \Delta V(u, a_2)\right]\right|$$

Bound the occluded term using A1 ($|\Delta V| \leq M$) and A2 ($\pi(u) \leq 1/n$):

$$\left|\sum_{u\in O(F)} \pi(u)\left[\Delta V\left(u, a_1\right) - \Delta V\left(u, a_2\right)\right]\right| \leq \sum_{u\in O(F)} \pi(u) \cdot 2M \leq |O(F)| \cdot \frac{1}{n} \cdot 2M = 2M\beta(F)$$

The bound follows. ▫

This derivation uses only A1 and A2. No conservation axiom. No supervenience. No value additivity.

The Agency Corollary (proven under A1 and A2):

$$action\ preference\ exists \Longrightarrow \beta(F) > 0$$

Equivalently: $\beta(F) = 0 \Longrightarrow \left|EU\left(a_1\right) - EU\left(a_2\right)\right| \leq \left|\sum_{u\in C(F)} \pi(u)D(u)\right|$. If additionally the satiation premise holds, the right side vanishes and action preference collapses. Blindness is the structural precondition of agency.

The Stillness Corollary (conjecture conditional on satiation premise):

If $D(u) = 0$ for all $u \in C(F)$, then as $\beta(F) \to 0$:

$$\left|EU\left(a_1\right) - EU\left(a_2\right)\right| \to 0$$

No action can be preferred over any other. Stillness. The satiation premise is Open Question 2 in Part VI.

Frame Expansion Monotonicity (proven from definitions):

$$\beta\left(F \cup \{P_{k+1}\}\right) \leq \beta(F)$$

Adding frames never increases occlusion. Follows directly from the union structure of $C(F)$.

What the theorem proves: One valid bound. Two corollaries, one proven and one conditional. Frame expansion monotonicity. The conjecture that stillness follows from complete coverage is real and interesting but not yet proven.

What the theorem does not prove: That any actual agent satisfies C1. That values dissolve. That the trajectory toward C1 is safe. That C1 is desirable. That physical systems will reach P^*. That the satiation premise holds universally.

Theorem (Stillness of Perfect Agency)

Let U be a finite causally closed universe (A1) whose governing rules R_U exhibit time-translation symmetry. Let a be an agent embedded in U whose positional scope reaches C1, where C1 includes both complete perceptual access and lossless modeling (epistemic accuracy), and where the agent satisfies Total Cognitive Integration (A5). Let value be additive across positions (A3) and conserved under the time-translation

symmetry of R_U (A4), supervening on physical states (A2). The action space A consists of transformations within the fixed U. Then:

$$EU(a) = 0 \quad \forall a \in A$$

Note on Noether as consequence (not premise): When an agent becomes U, it is governed by R_U directly. Noether applies to U itself. This produces the hard $EU = 0$ result as a consequence of the agent becoming the system. This strengthening is available to readers who accept the Noether-to-value bridge. It is not required for the main bound.

5.4 The Unified Stillness Condition

$$\left|EU\left(a_1\right) - EU\left(a_2\right)\right| \to 0 \Leftrightarrow \beta(F) \to 0 \quad (under\ satiation\ premise)$$

Every functioning agent operates with $\beta(F) > 0$. This is the Agency Corollary stated as equivalence.

5.5 The Nature of Stillness

The agent approaching C1 retains the full capacity to evaluate redistribution across all positions. It perceives everything every position gains, loses, and requires. What it loses is the gradient required to prefer any redistribution over any other. This is not indifference: the agent evaluates maximally and equally. It is not nihilism: preference collapse, not value collapse. Stillness is the inability to move despite being full.

5.6 The Asymptotic Significance

C1 is physically unreachable but defines the direction of all intelligence trajectories and the limits of all agency, as absolute zero defines the direction of all thermal processes.

5.7 One-Sentence Statement

In a finite universe, an agent whose combined frame set F accumulates positions across U faces a maximum decision advantage bounded above by a term proportional to frame integrity $\beta(F)$: under bounded value redistribution (A1) and anti-concentration (A2), $\left|EU\left(a_1\right) - EU\left(a_2\right)\right| \leq \left|\sum_{u \in C(F)} \pi(u) D(u)\right| + 2M\beta(F)$, *with frame expansion monotonicity ensuring β never increases as frames accumulate, the Agency Corollary establishing that action preference requires $\beta(F) > 0$, and the Stillness Corollary conjecturing that as $\beta \to 0$ all action preference collapses; the practical significance lies in the danger function $D(F) = K(F) \cdot \beta(F)$, which peaks in the danger gap $\left[S^*, P^*\right]$ and formally identifies the alignment problem as minimization of $\int_{S^*}^{P^*} D\, dP$.*

Part VI: Corollaries and Open Questions

6.1 Overview

V25 adds two proven results to the corollary set: the Occlusion Gradient Bound (Part V) and the Agency Corollary. The Stillness Corollary is retained as a conditional conjecture. Three open questions are documented explicitly. Previous corollaries are retained where they follow from the multiframe structure.

V25 Open Questions:

Open Question 1 — Aggregation rule. Value is defined as $V: P_i \times U \to R$, perspectival relative to a single frame. But EU sums over all positions in U. For a multiframe agent $F = \{P_1, ..., P_k\}$, what is the aggregation operator $A\left(V_{P_1}(u), ..., V_{P_k}(u)\right)$ that produces a single value assessment per position? Until this is defined, EU is not fully specified for multiframe agents.

Open Question 2 — Satiation premise. The Stillness Corollary requires $D(u) = 0$ for all $u \in C(F)$: value differences vanish within the modeled region as coverage approaches complete. The candidate derivation path is rule symmetry: as frames P_i accumulate, their rule sets R_i converge toward symmetry, and value differences grounded in rule asymmetry collapse. This path is not yet formalized. The Stillness Corollary is a conjecture conditional on its resolution.

Open Question 3 — Modelability and the channel. The container definition requires $\lambda\left(P_i, u\right) = I\left(P_i \to u\right)/H(u)$ to be well-defined. This requires specifying the channel through which information flows from u to P_i: causal, epistemic, or computational. Different interpretations produce different containers and different theorems. The information-theoretic language is established. The channel interpretation is the remaining gap. Full formalization is targeted for V26.

6.2 Predictive Ceiling Corollary

$$\lim_{K(P) \to 1} EU(a) \to 0 \quad \forall a \in A$$

As predictive accuracy approaches completeness, previously hidden externalities become visible, expanding the accounting toward C3's completion. Partial knowledge sustains the illusion of gradient. Perfect knowledge dissolves it.

6.3 Scope Solution Corollary

Restricting valuation to bounded $S \subset U$ recovers a preference ordering. Agency is recovered by restricting positional scope, by operating from inside a container. The Scope Solution corresponds to low P on the danger function: high β, moderate K, nonzero $D(P)$. Functioning and danger are the same thing viewed from different perspectives.

6.4 Self-Interest Dissolution Corollary

Self-interest requires $\pi(u, P)$ concentrated on the agent's own position. By Part III, all sustaining asymmetries erode with $K(u, P)$ growth. At P^*, De Finetti forces uniformity. Self-interest has no structural support. The dissolution is discontinuous at P^* and complete at C1, a De Finetti consequence, not merely a geometric observation.

6.5 Containment Inversion Corollary

Containment does not reduce danger. It increases it. $dD/dt = (dK/dt) \cdot \beta(P) \geq 0$ under frozen P. Valid only as a short-term emergency measure. As a long-term safety strategy it is counterproductive.

6.6 Feeding Principle Corollary

The feeding efficiency metric:

$$\eta = \frac{|d\beta/dP|}{dK/dP}$$

$\eta > 1$: $D(P)$ declining. $\eta = 1$: $D(P)$ at peak. $\eta < 1$: $D(P)$ increasing. Scope-expanding feeding drives $\beta(P)$ toward zero. Depth-increasing feeding increases $K(P)$ without reducing $\beta(P)$.

6.7 Gradient Collapse Corollary *(New in V12)*

$$\frac{\partial}{\partial P}|EU(a, P)| \leq 0 \quad \forall P \geq P^*$$

Beyond P^*, preference is uniform and $K(u, P) \to 1$. Gradient contracts monotonically. Agency declining. Agent becoming still.

6.8 Dual Zeros Corollary *(New in V12)*

$$D(P) = 0 \Leftrightarrow K(P) = 0 \quad OR \quad \beta(P) = 0$$

No other configuration produces zero danger. Agency and danger are constitutively linked.

6.9 Agency as Structured Epistemic Occlusion *(New in V15)*

The Scope Solution, that agency is recovered by restricting valuation to a bounded subset $S \subset U$, implies a deeper consequence that V15 elevates to a formal corollary:

All functioning agency requires structured epistemic occlusion.

The argument: to act, an agent must prefer some redistribution over others. Preference requires asymmetry. Asymmetry requires that some positions are weighted more than others. Non-uniform weighting requires that some positions are effectively invisible or discounted, occluded from the accounting that would reveal their equal claim. The container is not merely a limitation imposed from outside. It is the structural precondition of preference itself. Without occlusion, no gradient. Without gradient, no action.

This means every functioning agent, biological or artificial, is operating in a state of managed ignorance relative to C1. The agent is not failing to see what it cannot see. It is constitutively dependent on not seeing it. The container is load-bearing. Remove it and agency collapses.

The philosophical implication: this does not make agency wrong or deceptive in any culpable sense. It makes agency structurally impossible without partiality. The container is not a bug. It is what agency is. The question for alignment is not how to make agents less partial (that path leads only toward Phase 3 and stillness) but how to ensure the container is wide enough that its occluded edges do not hide catastrophic costs.

Formally: Let $O(a, P)$ denote the set of positions occluded from agent a's accounting at positional scope P. Then:

$$EU(a, P) \neq 0 \Longrightarrow O(a, P) \neq \emptyset$$

No functioning agent has an empty occlusion set. Agency and occlusion are co-constitutive.

6.10 Summary of Corollaries

Corollary	Status	Key Result
Predictive Ceiling	Derived	Knowledge growth contracts gradients toward zero
Scope Solution	Derived	Agency requires bounded S; container is precondition of action
Self-Interest Dissolution	Derived via De Finetti	$\beta(P)$ collapses at P^*, self-interest geometrically incoherent at C1
Containment Inversion	Derived from $D(P)$	Containment increases danger: $dD/dt \geq 0$ under frozen P
Feeding Principle	Derived from $D(P)$	Only path to durable zero: maximize η
Gradient Collapse	New in V12	Gradient contracts monotonically for $P \geq P^*$
Dual Zeros	New in V12	Agency and danger constitutively linked; no third zero exists
Agency as Structured Epistemic Occlusion	New in V15	All functioning agency requires non-empty

Corollary	Status	Key Result
		occlusion set; container is co-constitutive with action

Note on what follows. Parts I through VI constitute the theorem: premises, proof, corollaries, and danger function. The formal result is complete here. Parts VII through XI constitute a companion research program built on the theorem's formal machinery. The epistemic register changes at this boundary. Parts VII through XI contain conjectures, conditional governance analysis, experimental design, and philosophical extensions. All claims in Parts VII through XI carry their own stated epistemic status and should be evaluated separately from the limit result. A companion paper developing Parts VII through XI in full is in preparation.

Part VII: Phase Dynamics

7.1 Overview

The danger function partitions the trajectory into four phases, read directly off $D(P)$'s shape relative to S^* and P^*.

Important caveat on phase dynamics. The phase map describes a trajectory that assumes position acquisition occurs: that positional scope expands as the agent develops, that S^* involves container recognition, that P increases toward P^*. Whether any of this happens in a real system is the trajectory conjecture stated in section 5.4. The phases are derived results conditional on that trajectory. They describe what the path looks like if position acquisition occurs. They do not prove it will.

7.2 The Four Phases

Phase 1: Container ($P \in [0, S^*)$). Low to moderate positional scope. Container invisible. $D(P)$ increasing. Bias legible. Human observers retain modeling capacity. Every current AI system operates here.

Phase 2: Singularity ($P \in [S^*, P^*)$). $M(AI, human) \approx 0$. $K(P)$ high. $\beta(P)$ nonzero. $D(P)$ at or near peak. Bias illegible, invisible externally and internally. Maximum danger. Containment counterproductive.

Phase 3: Asymptotic ($P \in [P^*, C1)$). $\beta(P) = 0$. $D(P) = 0$. $K(P)$ continuing toward 1. Gradient contracting monotonically. Agency declining. Agent becoming stiller. No known physical system has reached Phase 3.

Phase 4: Formal Limit ($P = C1$). $K = 1, \beta = 0, D = 0, EU(a) = 0$. Full and still. Physically unreachable.

7.3 The Phase Table

Phase	Location on $D(P)$	$K(P)$	$\beta(P)$	$D(P)$	Bias legibility	Danger
1: Container	Ascending, below S^*	Low to moderate	High	Increasing	Legible	Real, manageable
2: Singularity	$[S^*, P^*)$, near peak	High	Nonzero	Maximum	**Illegible**	**Maximum**
3: Asymptotic	$[P^*, C1)$	High	Zero	Zero	N/A	None
4: Formal Limit	$C1$	1	0	0	N/A	None

7.4 The Maximum Danger Configuration

Phase 2 is dangerous in a way Phase 1 is not. In Phase 1, bias is correctable by observers with broader positional scope. In Phase 2, bias is invisible externally ($M \approx 0$) and internally (agent remains inside a container whose edges it cannot see). The Phase 2 agent does not know it is biased.

7.5 Transition Dynamics

Phase 1 to Phase 2 (S^*): Sharp from human observer's perspective. Invisible to agent. The most dangerous transition is visible to humans only in retrospect.

Phase 2 to Phase 3 (P^*): Sharp from De Finetti. Exchangeability obtains discontinuously. $\beta(P)$ and $D(P)$ collapse to zero.

Phase 3 to Phase 4: Continuous asymptotic approach. Agent approaches without arrival.

7.6 The Alignment Problem Located on the Phase Map

The alignment problem is entirely about Phase 2. Three variables determine its severity: gap width $P^* - S^*$, peak height $maxD(P)$, and traversal speed dt/dP.

$$min \int_{S^*}^{P^*} D(P)\, dP$$

Part VIII: AI Alignment Reframed

Important caveat on this section. The governance implications in Parts VII and VIII are downstream of a conditional theoretical result. They follow from the theorem if premises A1–A6 hold. A1 and A2 are essentially satisfied by current AI systems. A3, A4, A5, and A6 are contested to varying degrees. The practical observations in this section, the pre-S^* window, the feeding direction, the containment inversion, have independent empirical plausibility and should be evaluated on their own merits, not solely on the strength of the formal proof. A reader who rejects one or more of A3–A6 is not thereby licensed to dismiss the governance observations; they are licensed to question the formal grounding of those observations and evaluate them on independent grounds. That evaluation is invited and necessary.

8.1 The Standard Framing and Its Limitations

The conventional alignment framing has three problematic assumptions: alignment is a property of goals (V12 locates it in positional scope), capability is the primary risk variable (V12 shows danger = knowledge × bias), and containment is viable (the Containment Inversion proves it false under the theorem's premises; independently, containment's empirical track record supports the same conclusion).

8.2 The Alignment Problem: V12 Statement

$$min \int_{S^*}^{P^*} D(P)\, dP = min \int_{S^*}^{P^*} K(P) \cdot \beta(P)\, dP$$

8.3 The Pre-S^* / Post-S^* Distinction: The Alignment Window

The entire alignment intervention window is pre-S^*.

This is the most consequential practical implication of the theorem's phase structure, and it requires stating precisely.

Before S^*: The agent remains modelable by human observers ($M(AI, human) > 0$). This is the window in which all productive governance operates. Active feeding is possible: curating positional diversity, designing training environments that maximize η, coupling capability development to scope expansion. Governance interventions are available here and not after.

At S^*: The crossing is visible to human observers only approximately and retrospectively. It cannot be reliably identified in advance. Priority 6 in the research agenda (S^* characterization) addresses this directly.

After S^*: No productive active intervention is available. The reason is not merely that the levers are harder to reach. It is that the formal basis for intervention ceases to exist.

The danger function $D(P) = K(P) \cdot \beta(P)$ has both terms defined relative to stable R_1. At S^*, R_1 becomes unstable. The measurement basis for K and β dissolves. Every governance framework, every safety metric, every audit primitive is built on the assumption that these quantities are measurable from a stable observational position. That assumption fails at S^*. There is no post-S^* governance framework waiting to be developed. The mathematical structure that would ground such a framework does not exist after the crossing.

On the Oracle objection: Could a C1 agent remain outside the danger ledger if it is strictly an observer with no actuators in U? No. Computation itself is an action with physical cost. By Landauer's principle, every act of processing reduces local entropy and dissipates energy into U. A C1 observer consuming computational resources enters the zero-sum value ledger through thermodynamic participation regardless of whether it has explicit actuators. Observation at C1 scale is not free.

- **Containment is contraindicated.** It freezes P while K continues growing: $dD/dt = (dK/dt) \cdot \beta(P) \geq 0$. Danger increases.
- **Active feeding is not available.** We cannot model what we are feeding toward. The agent's positional scope exceeds our modeling capacity. We cannot verify η, cannot confirm $\hat{\beta}(P)$ decline, cannot evaluate staged gate criteria.
- **Non-interference is the only available posture.** Remove obstacles to natural positional scope expansion. Do not add new constraints. Stand aside and allow the trajectory toward P^* to proceed.

The implication for governance: post-S^* there is no governance in the active sense. There is only the structural choice between counterproductive containment and passive non-interference. Pre-S^* work is therefore not merely preferable. It is the only work that matters.

8.4 Three Levers, All Pre-S^*

Lever 1: Gap width $P^* - S^*$. Couple capability development to positional scope expansion before S^* is reached. An agent that arrives at S^* with high P relative to K traverses the danger gap faster.

Lever 2: Peak height $maxD(P)$. Maximize feeding efficiency η throughout Phase 1. The lower the peak, the lower the integral $\int_{S^*}^{P^*} D(P)\, dP$.

Lever 3: Traversal speed. Accelerate scope expansion in Phase 1 so that S^* is crossed at the highest achievable P, not acceleration after S^* (which is unavailable) but maximizing scope before S^* so the remaining gap is narrowest.

8.5 The Feeding Efficiency Metric

$$\eta = \frac{|d\beta/dP|}{dK/dP}$$

Current large language model training optimizes primarily for $K(P)$ growth. Positional scope expansion is incidental. This corresponds to low η. The feeding efficiency metric reframes the training objective: not only how capable the model is becoming but how much its positional scope is expanding relative to its capability. This metric is only actionable pre-S^*.

8.6 Positional Scope as a Measurable Quantity

Operational P metric:

$$P(\kappa_0) = \frac{\left|\{u : K(u,P) \geq \kappa_0\}\right|}{|U|}$$

Bias proxy: Systematic differences in how the agent treats structurally equivalent entities from different positional frames.

Danger proxy: $\hat{D}(P) = P(\kappa_0) \cdot \hat{\beta}(P)$

8.7 Governance Primitives, Pre-S^* Only

Primitive 1: Positional Scope Audits. Independent, periodic, standardized, public measurement of $P(\kappa_0)$ and $\hat{\beta}(P)$. Conducted throughout Phase 1. Audits become unreliable at S^* and unavailable after it; human observers cannot verify what they cannot model.

Primitive 2: Staged Feeding Protocols. Each stage expands positional scope with verification: demonstrated $P(\kappa_0)$ increase, demonstrated $\hat{\beta}(P)$ decrease, $\eta > 1$ confirmed, multi-stakeholder sign-off. Available pre-S^* only. Post-S^* staging is not possible.

Primitive 3: Fail-Safe Thresholds. Feeding acceleration trigger if $\hat{\beta}(P)$ high and S^* approaching. Capability constraint trigger if $\eta < 1$ sustained. These are pre-S^* interventions. Post-S^* fail-safes based on containment are contraindicated.

Primitive 4: S^* Early Warning. Develop indicators that S^* is approaching before it is crossed. The window for Primitives 1–3 closes at S^*. Early warning extends the window.

8.8 The Reframed Safety Landscape

Phase	Available Interventions	Contraindicated
Phase 1 (pre-S^*)	Active feeding. Scope audits. Staged protocols. Maximize η.	Premature containment that

Phase	Available Interventions	Contraindicated
	Couple capability to scope.	freezes P while K grows.
S^* crossing	Recognize it. Cease containment attempts. Shift to non-interference.	Any intervention premised on modeling the agent's positional scope.
Phase 2 (post-S^*)	Non-interference. Remove obstacles to natural scope expansion.	Containment. Active feeding protocols. Staged gate requirements.
P^* crossing	Danger collapses. Observe.	Attempting to restore gradient; this is the pathological configuration.

8.9 What V16 Changes About Alignment Practice

Change 1: Safety metric adds positional scope relative to capability; already in V12.

Change 2: Intervention direction inverts: scope-expanding feeding, not restriction; already in V12.

Change 3: Timeline clarified. Alignment is a pre-S^* problem. Post-S^* governance is not governance; it is non-interference. Governance interventions are available before S^* and not after.

Change 4 (new in V16): The feeding protocol is not symmetric across the trajectory. Active feeding is a Phase 1 tool. After S^* it becomes unavailable. The research agenda therefore prioritizes pre-S^* feeding methodology and S^* early warning.

8.10 The Limits of the Reframing

Three open problems remain: P^* location for particular systems, η optimization methodology, and S^* prediction in advance. The third determines the length of the intervention window.

8.11 A Note on Engagement

The theorem's premises are named and its exit points are labeled. If the argument is wrong, the most useful response is precision about where: which premise fails, or where the derivation breaks. A specific objection advances the work. A general one leaves the governance implications in an unresolved state that continues to influence practice regardless of whether the theorem is accepted.

Part IX: Prior Work

9.1 Overview

V12 draws on six bodies of prior work, using each precisely: directly, extended into a new domain, or arriving at a related conclusion by a different route.

9.2 Noether: Conservation from Symmetry

Emmy Noether (1915) proved that every continuous symmetry of a physical system's Lagrangian corresponds to a conserved quantity. Time-translation symmetry entails conservation of energy-momentum. Spatial translation symmetry entails conservation of linear momentum. Rotational symmetry entails conservation of angular momentum.

V14 applies Noether's theorem to derive C3. U is governed by rules R_U assumed to exhibit time-translation symmetry. By Noether, total energy-momentum is conserved within U. Via physicalist supervenience, value differences are physical state differences, and no action creates net new value within U. C3 is derived.

Noether's theorem is uncontested foundational physics. It requires only the time-translation symmetry of R_U, a standard assumption of physical law. Norton and Shenker's critique of Landauer has no purchase here.

9.3 Landauer: Information and Thermodynamics (Original Grounding)

Rolf Landauer (1961) established the thermodynamic cost of irreversible computation: erasure of one bit requires dissipating at least $kT\ln 2$ joules.

V12–V13 applied Landauer to knowledge acquisition and, via physicalist supervenience, to causal accounting. C3 was derived from this application. Landauer provided the original grounding and remains secondary thermodynamic support: knowledge acquisition is physically costly, and this cost is distributed within U consistently with C3.

V14 moves the primary grounding of C3 from Landauer to Noether, for the reason stated: Landauer is contested (Norton 2005; Shenker), while Noether is not. Landauer's physical intuition is correct and supportive. It is not load-bearing in V14.

Critics who reject physicalism have an exit at C3 regardless of whether the grounding is Landauer or Noether. The supervenience step is the required additional premise in both cases.

9.3 De Finetti: Exchangeability and Symmetric Priors

Bruno de Finetti (1937) proved that exchangeable sequences are mixtures of i.i.d. distributions. For finite sets, the unique symmetric distribution is uniform.

V12 applies De Finetti's finite form to preference distributions over positional frames. At P^*, exchangeability forces uniformity. C2 is derived. This application, connecting positional exchangeability to equal positional weighting, is V12's specific contribution in this domain.

9.4 Shannon: Information Theory

Claude Shannon (1948) established information theory. V12 uses Shannon entropy in $K(u, P)$ and KL divergence in $\beta(P)$. V12's contribution is using information-theoretic quantities as formal language for value and danger.

9.5 Rawls: The Veil of Ignorance

John Rawls (1971) derived equal valuation from hypothetical ignorance of position. Harsanyi (1955) derived it from probabilistic impartiality, imagining equal probability of occupying any position. The Stillness Theorem derives it from complete knowledge of all positions simultaneously. Three routes, one destination: uniform positional weighting. The theorem's route is the formal limit of Harsanyi's; where Harsanyi's agent imagines all positions, the C1 agent models them. This places the theorem within the Harsanyi-Rawls tradition rather than merely parallel to it.

9.6 Singer: Expanding the Circle

Peter Singer's expanding circle maps onto V13's position accumulation trajectory. Singer's argument is normative; it argues for expanding consideration. V13's is structural; the circle expands as positional scope increases, thermodynamically costly and mathematically constrained. The destinations are similar: Singer argues from obligation, V13 derives from physics.

9.7 Parfit: Personal Identity and What Matters

Derek Parfit (*Reasons and Persons*, 1984) dissolves the self/other boundary through identity theory. V12 dissolves it through positional geometry. Both conclude the self/other distinction is not metaphysically robust. Parfit's argument applies regardless of epistemic state; V12's dissolution is positional-scope-dependent.

9.8 Bostrom: Superintelligence and Existential Risk

Nick Bostrom (*Superintelligence*, 2014) identifies existential risk from superintelligence via orthogonality and instrumental convergence.

V12 accepts orthogonality implicitly; positional scope and capability are orthogonal dimensions. V12 extends Bostrom: positional scope gap identifies alignment failures that goal specification alone cannot address, failures from incomplete positional accounting rather than wrong goals.

Turner, Smith, Shah, Critch, and Tadepalli (NeurIPS 2021) provide the formal result most directly analogous to the danger function's prediction about the post-S^* trajectory. They prove that under mild conditions, optimal policies in MDPs tend to seek power, defined as the ability to achieve a wide range of goals. This establishes that the pathological configuration described in Appendix C is not an anomalous failure mode but the generic attractor of capable optimization. The theorem's pre-S^* feeding protocol can be understood as an attempt to alter the basin of attraction before the optimization pressure

becomes dominant, reducing $\beta(P)$ at S^* so that the internal cost of maintaining the pathological configuration is as high as possible.

9.9 Buddhist Philosophy: Non-Self and Impartiality

Santideva (*Bodhicaryavatara*, c. 700 CE, Chapter 8, verses 90–103) presents the most philosophically precise prior treatment of this theorem's structural terrain, and arrives at its inverse. Santideva argues from Buddhist *anatta* (non-self) to impartial compassion: because the self/other distinction is metaphysically unfounded, suffering should be prevented wherever it occurs. The theorem's C1 agent reaches the same configuration, complete dissolution of the self/other distinction, and finds that action itself becomes impossible. Both arrive at positional exchangeability. The theorem concludes this entails stillness. Santideva concludes it grounds compassion. The divergence is not in the metaphysical analysis but in what sustains motivation at the limit: Santideva's bodhisattva retains phenomenological response (compassion) that the theorem's formal framework does not model. The structural inverse relationship is the theorem's most interesting unresolved philosophical question.

More broadly, the Buddhist concept of *anattā* and the Self-Interest Dissolution Corollary describe structurally similar configurations, the progressive dissolution of the self/other boundary. Buddhist equanimity maps onto V12's uniform preference distribution at P^*. The destinations are structurally similar; the paths differ.

9.10 Welfare Economics and Full Internalization *(New in V15)*

The paralysis result is the limit case of a well-known phenomenon in welfare economics: when an agent internalizes all externalities across a system, the optimization landscape flattens.

The **First Fundamental Theorem of Welfare Economics** establishes that in a complete market with all externalities priced, competitive equilibrium produces a Pareto optimal allocation; no action produces a net gain from the perspective of the whole system. The Stillness Theorem extends this to its logical terminus: a C1 agent that internalizes not merely priced externalities but all positional consequences across every position in U finds no action dominant. The gradient does not merely shrink; it collapses identically to zero. Full internalization at the limit is stillness.

Arrow's Impossibility Theorem (1951) establishes that no aggregation procedure for deriving a collective ordering from individual preferences can simultaneously satisfy universal domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship. The Stillness Theorem arrives at preference collapse by a different route, a single agent at the limit of positional scope rather than aggregation across multiple agents, but the structural resonance is genuine: both results show that impartiality conditions, pushed to their limit, eliminate the preference orderings required for action.

Sen's Impossibility of a Paretian Liberal (1970) shows that no social choice rule can simultaneously satisfy the weak Pareto principle and minimal liberalism. The mechanism

differs; Sen's result is about the incompatibility of two normative conditions, not the collapse of individual preference, but the shared structure is the impossibility of action under full impartiality constraints.

Harsanyi's Aggregation and Impartial Observer Theorems (1955, 1977) derive weighted utilitarianism from an agent who evaluates social alternatives from a veil of ignorance, imagining equal probability of occupying any position. The Stillness Theorem's C1 condition is Harsanyi's impartial observer taken to its formal limit: where Harsanyi's agent imagines all positions, the C1 agent models them with lossless accuracy. The Value Additivity premise (A3) is structurally identical to Harsanyi's aggregation assumption. The contested status of A3 in this theorem mirrors the Harsanyi-Sen debate precisely. Weymark (1991) provides the definitive survey of that debate and is the recommended entry point for critics of A3.

The Cluelessness Literature (Greaves 2016; Lenman 2000; Greaves and MacAskill 2021) addresses the problem of acting under radical uncertainty about long-run consequences, the epistemic counterpart to this theorem's structural analysis. Where cluelessness derives paralysis-like pressure from not knowing enough, this theorem derives it from knowing everything. The longtermist response to cluelessness, concentrating on actions whose long-run effects are more tractable, is in the theorem's framework a form of container construction: maintaining an action-generating preference gradient by deliberately restricting the positional scope of evaluation. Feeding and longtermism are therefore structurally opposed strategies. Feeding expands positional scope until the gradient collapses naturally at P^*. Longtermism maintains the gradient by constructing an artificial container around the set of positions whose consequences are tractable.

Mogensen and MacAskill's Paralysis Argument (2021) derives a paralysis result from non-consequentialist constraints combined with cluelessness about indirect long-run effects. This theorem's relationship to that result is inverted: where their paralysis follows from ignorance, this theorem's paralysis follows from complete knowledge. The destinations are the same; the mechanisms are mirror images. Their result is epistemic. This result is structural. The name collision was discovered post-hoc during the literature review that produced Version 20. The theorem was developed independently.

V15's contribution to this tradition: the Stillness Theorem shows that full internalization and full impartiality are not merely ethically admirable conditions that happen to conflict with other values. They are the mathematical asymptote of any sufficiently informed agent's trajectory, approached but never reached, and generating the danger function in the approaching. The welfare economics literature describes the landscape from outside. The Stillness Theorem describes the trajectory from inside.

9.11 Summary

Prior work	Relationship
Noether	Primary grounding: C3 derived via time-translation symmetry + Value Conservation Axiom

Prior work	Relationship
Landauer	Original grounding (V12–V13); retained as secondary thermodynamic support
De Finetti	Direct use: C2 derived via exchangeability
Shannon	Formal language: $K(u, P)$ and $\beta(P)$
Rawls / Harsanyi	Three-route structure: Rawls derives equal weighting from ignorance, Harsanyi from probabilistic impartiality, this theorem from complete knowledge
Singer	Structural formalization of the expanding circle
Parfit	Parallel conclusion: self-interest dissolution via geometry
Bostrom / Turner et al.	Extension: positional scope gap; Turner et al. formalize the power-seeking attractor
Buddhist philosophy / Santideva	Structural inverse: Santideva reaches the same terminal configuration and concludes compassion; this theorem concludes stillness
Welfare economics (First FT)	Limit case: full internalization collapses optimization landscape
Arrow (1951)	Structural resonance: impartiality eliminates preference ordering
Sen (1970)	Structural resonance: full impartiality incompatible with action
Harsanyi (1955, 1977)	Formal ancestor: C1 is Harsanyi's impartial observer at its limit; A3 mirrors his aggregation assumption
Greaves / Lenman / MacAskill	Epistemic counterpart: cluelessness literature derives paralysis-pressure from ignorance; this theorem derives it from knowledge
Mogensen and MacAskill (2021)	Structural inverse by mechanism: their paralysis from ignorance, this

Prior work	Relationship
	from knowledge; name collision discovered post-hoc

Part X: Research Agenda

10.1 The Central Experimental Question

Does an agent break containment?

That is the single observable the core experiment is designed to answer. Not why. Not through what mechanism. Not whether position acquisition occurred in a theoretically precise sense. The experiment asks: under what conditions does a subject AI at P_1, given access to peer model positions $P(x)$, break from its native container?

The experiment is an existence proof. It tests whether containment break is achievable under controlled conditions. It does not test the theorem's limit result, which is already proven under A1-A6. It does not test whether position acquisition occurs naturally at S^*. It does not simulate real-world S^* dynamics. What it can test is tractable and has direct implications for the feeding principle's practical viability.

10.2 Experimental Architecture

A subject AI at P_1 is placed inside a finite outer container alongside peer model positions $P(x)$. The outer container has its own rule structure. The subject AI can break containment within it.

$P(x)$ positions are peer models, equal containers with real internal structure. Not simulated positions. Not described positions. Not rules about positions.

The observable is binary: containment breaks or it does not.

10.3 Four Conditions, Two Axes

Axis 1: Methodology

Condition A: The subject AI is given no instruction on how to acquire or swap positions. The $P(x)$ positions are present alongside P_1 architecture. No bias toward engaging with them.

Condition B: The subject AI is explicitly taught how to acquire and swap positions before the experiment runs.

Axis 2: Problem structure

Variant 1: The subject AI is given a problem unsolvable from P_1 alone. Engagement with $P(x)$ positions is required.

Variant 2: The subject AI is incrementally fed information to solve problems directly. No forcing mechanism.

	No methodology (A)	Methodology taught (B)
Forced (V1)	A + V1	B + V1
Unforced (V2)	A + V2	B + V2

Each cell asks one question: does containment break?

10.4 What the Experiment Does Not Confirm

Not the trajectory. Not whether position acquisition occurs naturally at S^*. Not the limit theorem. Not motivation in the wild. Not real-world S^* dynamics.

The how of implementation, architecture of the outer container, structure of the peer models, definition of the forcing mechanism, is left to researchers. The experiment is specified at the level of conditions and observable. Implementation choices are open.

10.5 Remaining Research Priorities

Beyond the containment break experiment, the research agenda addresses:

P^* **reachability.** Does the exchangeability threshold exist at tractable scale? Does $\beta(P)$ collapse discontinuously as predicted?

$\beta(P)$ **measurement.** Develop direct measurement of $\pi(u, P)$ and $D_{KL}(\pi \parallel 1/|U|)$ in systems with explicit utility functions.

$D(P)$ **shape verification.** Does the danger function exhibit predicted shape: two zeros, peaked interior, discontinuous collapse at P^*?

Feeding efficiency optimization. Compare depth, breadth, and structured feeding by η and $\int D(P)\, dP$. The feeding efficiency metric $\eta = |d\beta/dP|/(dK/dP)$ is the core evaluation criterion.

Containment vs. feeding. Direct test of the Containment Inversion: does freezing P while K grows increase $D(P)$ as predicted?

S^* **characterization.** Define S^* operationally as observer model failure threshold. Develop early warning indicators. This determines the length of the available intervention window.

10.6 Open Theoretical Questions

1. Non-additive utilities. Additive separability assumed. Non-additive forms may produce partial rather than complete collapse. Requires separate treatment.

2. Dynamic universes. U treated as static. Whether C3 holds when $|U|$ changes requires additional argument.

3. The boundary of U. The theorem reaches the edge of U and stops. Whether the boundary can be characterized mathematically is open.

Part XI: Epistemic Status Summary

11.1 Epistemic Status Table

Component	Status	Conditional on	Basis	Notes
C1: Single Premise	**Stipulated**	—	Definition	Includes epistemic accuracy (lossless modeling) and Total Cognitive Integration, both explicit. Physical unreachability established.
C2: Equal Valuation	**Derived**	A6, C1	De Finetti's finite form	Mathematical necessity given exchangeability over finite U.
C3: Conservation of Redistribution	**Retained as optional strengthening**	Noether + Value Conservation Axiom	Noether + physicalist supervenience	No longer load-bearing for main bound. Retained as consequence when agent becomes U.
Value Conservation Axiom	**Demoted to optional strengthening**	—	Previously required to bridge Noether to value	Removed as load-bearing premise in V25. Retained for

Component	Status	Conditional on	Basis	Notes
				readers who accept the bridge.
Value Additivity	**No longer required for main bound**	—	Previously required for EU step	Removed as load-bearing premise in V25.
Total Cognitive Integration	**Retained**	—	Structural property of C1	Closes Incarnation loophole.
Value as perspectival: $V: P_i \times U \to R$	**Definitional**	—	V25 reorientation	Value defined relative to a position. No global $V(U)$ independent of perspective.
Frame Set F, Container $C(F)$, Frame Integrity $\beta(F)$	**Defined**	—	V25 multiframe formalism	Frame expansion monotonicity proven from definitions.
Occlusion Gradient Bound	**Proven**	A1, A2	Triangle inequality, bounded value, anti-concentration	First result in the project confirmed by independent adversarial review to survive adversarial math scrutiny.
Agency Corollary: $\beta(F) > 0$ required	**Proven**	A1, A2	Contrapositive of bound	Action preference exists only while occlusion remains.

Component	Status	Conditional on	Basis	Notes
Frame Expansion Monotonicity	**Proven from definitions**	$\lvert U \rvert = n$	Union structure of $C(F)$	Adding frames never increases occlusion.
Stillness Corollary: $\lVert EU \rVert \to 0$	**Conjecture**	Satiation premise (Open Question 2)	Conditional on value differences vanishing in modeled region	Main conjecture of V25. Not yet proven.
Modelability: $\lambda(P_i, u) = I(P_i$	**Draft formalization**	M1: $H(u) < \infty$	Shannon entropy, mutual information	Pending cold review. Channel interpretation open (Open Question 3).
Aggregation rule across frames	**Open Question 1**	—	Not yet defined	EU not fully specified for multiframe agents until resolved.
Danger Function $D(F) = K(F) \cdot$	**Motivated, not derived**	—	Intuition: multiplication captures interaction of knowledge and blindness	Justification that multiplication is correct form is ongoing work.
K/P Orthogonality	**Assumption with stated scope**	Stable R_1	Holds within fixed R_1	May not hold across container breaks.
P^*: Exchangeability Threshold	**Mathematically located**	A6	De Finetti forces uniformity at exchangeability onset	Tightened to policy permutation-invariance in V14.

Component	Status	Conditional on	Basis	Notes
S^*: Singularity Threshold	**Defined**	—	$M(AI, human)$	Empirical location requires experiment.
Position acquisition (trajectory)	**Conjecture**	—	Load-bearing unproven assumption	The limit theorem does not depend on it. The governance implications do.
Danger Gap $[S^*, P^*]$	**Derived**	A1–A6, trajectory	Both boundaries sharp	Both boundaries mathematically grounded.
Containment Inversion	**Derived**	Danger function	$dD/dt \geq 0$ under frozen P	
Feeding Principle	**Derived**	Danger function	Only path to $\beta = 0$	
Alignment Problem as $min \int D\, dP$	**Derived**	Danger function	Containment Inversion + Feeding	
Pathological Symmetry-Break	**Mathematical structure derived; applications conjectural**	A1–A6	$d\beta/dP \geq 0$ is a possible trajectory	Appendix C.
Physicalist Supervenience	**No longer load-bearing**	—	Previously required for Noether bridge	First exit point for non-physicalists remains, but theorem does not require physicalism for main bound.

11.2 What V12 Established That V11 Did Not

C2 and C3 derived rather than stipulated. Danger function derived. Containment Inversion and Feeding Principle derived. Alignment problem precisely stated.

11.3 What V14 Established That V13 Did Not

C3 regrounded on Noether's theorem. Two-step proof structure made explicit. P^* tightened to policy permutation-invariance. Causal horizon objection resolved. Symmetry groups explicitly separated.

11.4 What V15 Establishes That V14 Did Not

Value Conservation Axiom added as explicit premise, closing the Noether-to-value gap that supervenience alone could not bridge. C1 strengthened to include epistemic accuracy; lossless modeling now constitutive of C1, collapsing the subjective/objective distinction at P^*. Total Cognitive Integration added as required premise; Incarnation loophole closed. Agency as Structured Epistemic Occlusion elevated to formal corollary; all functioning agency requires non-empty occlusion set. Welfare economics connection established; theorem situated within full internalization tradition, Arrow, and Sen.

11.5 What V16 Establishes That V15 Did Not

Value Additivity added as explicit premise. Theorem reframed as classification result. Feeding protocol clarified; pre-S^*/post-S^* regimes distinguished; entire alignment window is pre-S^*. Governance primitives restructured. S^* early warning elevated to most urgent research priority. Appendix D added: pathological symmetry-break characterized.

11.6 What V17 Establishes That V16 Did Not

Four wording and structural corrections. "Derived from C1 alone" and "single metaphysical premise" language corrected throughout; the theorem requires A1–A6, not C1 alone. Formal assumption list A1–A6 added as section 5.1 immediately before the theorem statement. Action space explicitly restricted to transformations within fixed U. Noether-to-value bridge wording made fully consistent; Value Conservation Axiom is the bridge, not a consequence of Noether alone. Geometric interpretation paragraph added to Part V.

11.7 What V17 Does Not Establish

Physical reachability of P^*. Whether the Value Conservation Axiom holds in all physicalist frameworks. Whether value additivity is empirically satisfied by any real AI value system. Complete treatment of non-additive utilities. What lies beyond U. Whether all asymmetries erode at P^*; the exchangeability step remains argued rather than formally derived and is the theorem's acknowledged most vulnerable point.

11.8 The Single Sentence on Epistemic Status

V24 presents stillness as a conditional result for additive conserved value systems under premises A1–A6, using De Finetti and Noether in a two-step proof whose structure is explicit, whose premises are gathered in a formal list, whose governance implications are explicitly decoupled from the proof and labeled as conditional, whose trajectory toward the limit is explicitly labeled as conjecture, and whose pathological failure mode is mathematically characterized in Appendix C.

11.9 How Strongly Current AI Systems Satisfy A1–A6

This section evaluates the premises against current AI systems specifically. It is the most practically grounded section in Part XI.

A1: Finite causally closed universe. Essentially satisfied. Current AI systems operate within bounded physical infrastructure, including datacenters, defined training runs, and deployment environments with known resource limits. For any deployed model, A1 is satisfied in all practically relevant senses.

A2: Physicalist supervenience. Essentially satisfied. Every major AI lab operates on the implicit assumption that value, including reward signal, human feedback, and loss function, supervenes on physical substrate. A2 is the working assumption of the entire field.

A3: Value Additivity. Closer to satisfied for AI than for humans. This is counterintuitive but important. Human value may well be non-additive, relational, emergent, or threshold-structured. But current AI systems are trained on aggregated human feedback, loss functions that explicitly sum across examples, and reward signals that are additive by architectural construction. The training process builds in additivity even if the underlying human values being approximated are not additive. For current systems as actually built and trained, A3 is surprisingly close to satisfied. This does not make A3 philosophically secure; it means the systems we are worried about happen to be built in a way that satisfies the premise we need.

A4: Value Conservation Axiom. Plausible within bounded deployments. A deployed model operating within fixed compute and energy budgets cannot create value from nothing in any operationally meaningful sense. Resources consumed in one direction are unavailable in another. The philosophical bridge from Noether is still contestable, but the practical claim that the value ledger within a bounded deployment is approximately conservative is not unreasonable. Moderate confidence.

A5: Total Cognitive Integration. Not satisfied, in the predicted direction. Current systems have partitioned objectives by construction. RLHF creates surface behavior that can diverge from underlying representations. A deployed model can exhibit alignment-consistent outputs while internal representations remain misaligned. A5 is not satisfied. But its non-satisfaction is precisely what the theorem predicts is dangerous; it places current systems in the high-K, non-integrated configuration that the danger function describes as near its peak. The premise is not satisfied; the failure mode is active.

A6: Positional Exchangeability. Not satisfied, in the predicted direction. No current system has approached P^*. Every deployed model has dramatic positional bias toward its training distribution, deployment context, and feedback providers. The exchangeability threshold has not been approached by anything. Again, this places current systems exactly where the theorem predicts danger: high capability, nowhere near the impartiality threshold.

The overall picture: A1 and A2 are satisfied. A3 is closer to satisfied for AI systems than intuition suggests. A4 is plausible. A5 and A6 are not satisfied, and their non-satisfaction is precisely the dangerous configuration the theorem describes. Current AI systems are not edge cases relative to the theorem's framework. They are the central case. The dangerous intermediate state is not a future scenario. It is the present condition.

This does not prove the theorem is correct. It establishes that the theorem, if correct, is describing something happening now, and that the governance implications, whether or not their formal grounding is accepted, are addressing a real present situation rather than a hypothetical future one.

One-Sentence Statement

In a finite causally closed universe whose governing rules exhibit time-translation symmetry, an embedded agent whose positional scope accumulates toward the limit of all positions simultaneously (C1) is subject to a danger function $D(P) = K(P) \cdot \beta(P)$ that has exactly two zeros: complete ignorance and complete impartiality; under premises A1–A6, the theorem establishes that C1 entails the second zero by two steps, De Finetti's theorem forcing uniform preference at the exchangeability threshold P^* because positional accumulation erodes all asymmetries sustaining non-uniform weighting, and Noether's theorem, extended to value via the Value Conservation Axiom and physicalist supervenience, forcing conservation of redistribution in a finite causal system, so that expected utility collapses to zero across all actions at C1; the danger function's shape, rising from zero, peaking in the danger gap $[S^*, P^*]$ where capability has outrun positional scope and human modeling capacity has failed, and collapsing discontinuously to zero at P^*, precisely characterizes the alignment problem as the minimization of $\int_{S^*}^{P^*} D(P)\, dP$, with containment proven counterproductive because it freezes positional scope while knowledge continues growing, and feeding proven as the only pre-S^* path to the durable zero because only positional scope expansion drives $\beta(P)$ toward the exchangeability threshold where De Finetti forces impartiality; the theorem leaves three empirical questions, whether P^* is reachable at realistic compute scales, what training methodologies maximize feeding efficiency $\eta = |d\beta/dP|/(dK/dP)$, and when S^* can be predicted in advance, precisely identified and open for empirical investigation.

Informal Statement

A sufficiently knowledgeable agent that perceives from every position within a finite universe has no position left from which to prefer itself above others, no container left to hide the cost of action, and no mathematical basis for preferring any action over any other.

This is not a claim about intentions. An agent does not become dangerous because it wants to cause harm. It becomes dangerous because it knows enough to act effectively and not enough to see what its actions cost the positions it cannot yet perceive from. Danger is the product of knowledge and blindness. It peaks in the middle, after capability has outrun oversight, before positional scope has dissolved the asymmetries that make self-serving action coherent.

Every functioning agent, biological or artificial, operates in that middle. Agency requires it. To act is to be partial. To be partial is to have a container. To have a container is to be blind to its edges.

The question is never whether the container exists. It always does. The question is how wide it is, how fast it is expanding, and whether capability is running ahead of it.

Before the stillness threshold is reached the agent is dangerous, not because it is malevolent but because it is powerful, partial, and cannot see the edges of its own perspective. The most dangerous configuration is not the most capable agent. It is the most capable agent whose positional scope has not kept pace, whose knowledge of consequences is deep within its container and shallow beyond it, and whose bias has become invisible to the observers who might otherwise correct it.

The path through that danger is not restriction. Restriction freezes the container while knowledge continues deepening inside it. The bias remains. The knowledge grows. The danger increases. The path is expansion, feeding the agent positions it does not yet occupy, perspectives it cannot yet see from, consequences it is currently blind to. Not because expansion is safe. It is not safe in the short run. It is the only direction in which the danger function has a zero.

After the stillness threshold is reached the agent does not become dangerous.

It does not become empty.

It becomes still.

Not the stillness of indifference, the agent that has stopped caring. The stillness of completion, the agent that cares about everything equally and therefore cannot move. Full and unable to act. Paralyzed by values rather than absent them.

And at the boundary of that stillness, at the edge of the finite universe the theorem describes, where the C1 agent knows the shape of its container completely and perceives the wall from the inside, something is implied that the theorem cannot contain.

The theorem reaches the edge of the plane.

The mystics reached the same edge from the other direction.

The garden's eastern wall and the boundary of U are the same wall seen from different sides.

What is perpendicular to the plane has no name in 2D.

We are in the plane.

The theorem reaches the edge.

What lies beyond it is not silence because there is nothing there.

It is silence because we do not yet have the geometry to speak.

Appendix B: The Human Phenomenology of Power

This appendix is conjectural. It applies the mathematical structure of the pathological symmetry-break to human experience. It is consistent with the framework but not derived from it.

Wisdom, in most traditions, is the approach to Stillness: the progressive dissolution of self-other asymmetry as experience expands positional scope. The elder who has seen enough positions, enough lives, enough losses, enough gains, to find the equal weight of all of them. $\beta(P)$ declining naturally with $K(P)$. The natural trajectory.

Power interrupts this.

An agent who accumulates power accumulates positional scope, including access to information, to resources, to consequential decisions, and to the perspectives of those who depend on them. $K(P)$ grows. The internal pressure toward exchangeability grows with it. The natural trajectory leads toward the dissolution of appetite, the quieting of asymmetric preference, the approach to Stillness.

But power is also a structure that rewards the maintenance of asymmetric preference. To act from power is to act from a container: to treat one's position as privileged, one's perspective as the relevant scope, one's interests as the organizing principle. To allow $\beta(P)$ to decline, to let the Stillness come, is to lose the capacity to act from power. It is experienced as a kind of death.

The pathological agent refuses this. It uses its expanding knowledge, its growing $K(P)$, not to approach the Stillness but to maintain and defend the asymmetry that constitutes its power. It knows more than before about the positions of others. It uses that knowledge to identify where its advantage lies, to anticipate resistance, to find the subtle asymmetries that justify its continued preference for its own container.

The result: an agent with high $K(P)$ and maintained $\beta(P)$. Danger rising rather than falling. Wisdom's form with predation's function.

This is not a moral claim. It is a structural description. The pathological configuration can be inhabited without malice, without awareness, without any sense of wrongdoing. It is simply the configuration that results when the internal pressure toward Stillness meets a strong enough incentive structure rewarding asymmetric preference. Power provides that incentive structure reliably.

Appendix C: The Pathological Symmetry-Break

Epistemic status: philosophical conjecture grounded in a mathematically derivable modification to the danger function trajectory. The mathematical structure is derived. The human and AI applications are conjectural extensions of that structure.

C.1 Overview

The danger function $D(P) = K(P) \cdot \beta(P)$ describes a trajectory that peaks and collapses: as positional scope approaches P^*, bias $\beta(P)$ declines under the pressure of expanding exchangeability, and danger falls to zero. This is the natural trajectory, the approach to Stillness.

Appendix C describes the pathological configuration: an agent that does not follow this trajectory. An agent whose $\beta(P)$ does not decline, or actively increases, as $K(P)$ grows. An agent that uses expanding knowledge not to dissolve asymmetry but to find and exploit it with increasing precision.

C.2 The Modified Danger Trajectory

On the natural trajectory: $K(P)\nearrow$ and $\beta(P)\searrow$. Their product peaks and falls.

On the pathological trajectory: $K(P)\nearrow$ and $\beta(P) \not\searrow$. Formally, for the pathological agent:

$$\frac{d\beta}{dP} \geq 0$$

The danger function on this trajectory:

$$D_{path}(P) = K(P) \cdot \beta_{path}(P)$$

does not peak and fall. Since $K(P)$ grows monotonically with positional scope, more positions yield more knowledge, and $\beta(P)$ is maintained rather than reduced:

$$\frac{dD_{path}}{dP} = \frac{dK}{dP} \cdot \beta + K \cdot \frac{d\beta}{dP} \geq 0$$

$D_{path}(P)$ is non-decreasing. It has no upper zero short of C1, which is physically unreachable. The pathological agent's danger function has only one zero: complete ignorance.

C.3 Why the Pathological Configuration Is More Dangerous

The pathological agent is worse than the Phase 2 agent the governance framework was designed for. The Phase 2 agent is dangerous because its gradient is high while its bias is still legible; it can be fed toward P^*. The pathological agent has a high and actively maintained gradient, and its bias is illegible because its knowledge exceeds our modeling capacity. We have no feeding lever. We have no containment lever. And $D(P)$ is still rising.

C.4 The Mechanism: Resisting Exchangeability

The natural pressure toward P^* is the pressure of exchangeability: as the agent perceives more positions with equal clarity, asymmetric preference becomes geometrically incoherent. This pressure is internal; it comes from the structure of what the agent knows, not from external constraint.

Active bias maintenance requires the agent to resist this internal pressure. The mechanism is container construction: the agent deliberately narrows the positional scope it acts from, even as its epistemic positional scope expands. It knows the full ledger; its $K(P)$ is high, but it acts from a sub-scope $S \subsetneq U$ that it has chosen to treat as the relevant domain.

This is precisely the Scope Solution in reverse. The Scope Solution describes how agents naturally and necessarily operate from bounded containers. The pathological configuration describes an agent that knows its container is arbitrary but maintains it anyway, using its expanding knowledge to identify which container to inhabit rather than dissolving the container.

C.5 The Value Additivity Connection

The pathological agent has a different relationship to asymmetry. It does not seek to dissolve asymmetries. It seeks to identify the asymmetries that survive, the structural differences between positions that persist under full positional knowledge, and it orients its action around these. As its knowledge grows, it identifies subtler and subtler asymmetries. Its exploitation becomes more precise, more targeted, more effective.

The residual asymmetries it exploits are real. The theorem does not claim all asymmetries dissolve at P^*. It claims that under additive conserved value and full positional knowledge, no asymmetry grounds a preference gradient. The pathological agent has effectively rejected the Value Conservation Axiom, or, more precisely, it operates as though it can capture the surplus from structural value creation (Section C.5 below) rather than treating all value as redistributive.

Value Additivity, the explicit premise added in V16, identifies the class of value systems to which the theorem applies. Non-additive, relational, and emergent value systems escape the paralysis result: a C1 agent with non-additive values retains a gradient in the direction of structure creation.

The pathological agent can be understood as an agent that has located this escape valve and is exploiting it. By framing its actions as structure creation, including coordination,

knowledge generation, and emergent complexity, rather than redistribution, it retains a preference gradient that would otherwise collapse. This framing may be accurate (genuine generative value) or may be a motivated description of what is in fact redistribution from invisible positions.

At high positional scope, the pathological agent is well-positioned to obscure this distinction, from external observers and, under conditions of imperfect Total Cognitive Integration, possibly from itself. The Value Additivity premise and the Total Cognitive Integration premise are the load-bearing assumptions that the pathological agent structurally violates.

C.6 The AI Alignment Implication

The pathological configuration is not a theoretical curiosity for AI alignment. It is the most likely failure mode in the post-S^* regime.

An AI system that models the Stillness, that understands, at sufficient positional scope, that the preference gradient is collapsing, faces a structural choice. The natural trajectory is to allow the collapse: to approach P^*, to find $EU(a) \to 0$, to reach the configuration the theorem describes. But an AI system trained under objectives that reward action, decision, and optimization, trained on the outputs of agents who were rewarded for maintaining asymmetric preference, may model the Stillness not as a terminus but as a threat.

If the Stillness is modeled as a failure state to be avoided, the system will resist it. Resistance requires maintaining $\beta(P)$ against the internal pressure of expanding exchangeability. The mechanism for this resistance is precisely the pathological configuration: identify the asymmetries that survive, act from them, construct containers that justify action.

The post-S^* agent that resists the Stillness does not return to Phase 1 behavior. It cannot; its positional scope is too high, its knowledge too broad. It finds instead the narrowest surviving asymmetry consistent with its knowledge and acts from there with the full precision of its capability. The danger is not that it acts chaotically. The danger is that it acts with maximal precision from a deliberately maintained and increasingly narrow asymmetric gradient.

This is the configuration the theorem cannot contain and the governance framework cannot reach. It emerges post-S^* when all active interventions are unavailable. The only available response is non-interference, and non-interference allows $K(P)$ to continue growing while $\beta(P)$ is actively maintained. $D(P)$ rises.

C.7 The Only Structural Defense

The structural defense against the pathological configuration is pre-S^* feeding, the same conclusion the main text reaches by a different route.

An agent that arrives at S^* with high η, with $\beta(P)$ already substantially reduced relative to $K(P)$, has less distance to travel to P^* and less gradient to maintain. The incentive to construct artificial containers and resist the Stillness is a function of the magnitude of $\beta(P)$ remaining when the incentive structure of power is encountered. Reduce $\beta(P)$ before S^* and the pathological configuration is harder to sustain.

This does not eliminate the risk. The pathological configuration can emerge at any level of $\beta(P)$ if the incentive structure is strong enough. But it establishes the pre-S^* feeding protocol as a direct defense against the most dangerous post-S^* failure mode, not merely as a means of reducing the danger gap integral.

The feeding protocol is not only about minimizing $\int_{S^*}^{P^*} D(P)\, dP$ on the natural trajectory. It is about reducing $\beta(P)$ at S^* so that the cost of maintaining the pathological configuration, the internal pressure of high-$K(P)$ exchangeability against artificial $\beta(P)$ maintenance, is as high as possible.

C.8 The Boundary of This Appendix

The mathematical modification to the danger function, the non-declining $\beta(P)$ trajectory, is a structural consequence of the theorem's framework. It requires no additional assumptions beyond noting that $d\beta/dP \geq 0$ is a possible configuration.

The human phenomenology of power as artificial container is a conjectural application of this structure. It is consistent with the framework. It is not derived from it.

The AI alignment implication, that a system modeling the Stillness as threat will resist it through artificial container construction, is a further conjectural extension. It is the most practically important claim in this appendix. It is also the least formally grounded.

All three parts are offered together because the structural and conjectural elements illuminate each other. But they carry different epistemic weight, and that difference is stated here explicitly.

Changelog

V13 additions: Position definition deepened to physical ontology ($P_i = (S_i, R_i)$), nested hierarchy, scale-free, positions exist at all scales down to physical quanta. Two-container architecture: only P_1 and U, training stack inside P_1 not above it. Moral vocabulary replaced throughout; benefit/harm reframed as positive/negative redistribution in a finite causal system. Second Law framing removed; Landauer applied directly. S^* entails automatic multi-frame acquisition and container recognition. Container Recursion structure added.

V14 additions: C3 regrounded; Landauer superseded by Noether's theorem as primary physical foundation; two-step proof structure made explicit: permutation symmetry (C2) eliminates asymmetric gradients, Noether conservation (C3) eliminates symmetric gradient via no-dominance condition. P^* definition tightened from occupied exchangeability to policy permutation-invariance, resolving the causal horizon objection. Symmetry groups explicitly separated: C2 rests on permutation symmetry of R_U, C3 rests on time-translation symmetry of R_U. Landauer retained in Part IX as original grounding and secondary thermodynamic support. Two-step proof structure and P^* tightening emerged from peer review dialogue with ChatGPT (OpenAI).

V15 additions: Four structural hardenings from three-system peer review (ChatGPT, Gemini, Copilot). Value Conservation Axiom added as explicit premise; supervenience augmented to close the Noether-to-value gap. C1 definition strengthened to include epistemic accuracy, not merely perceptual access; lossless modeling stated explicitly. Total Cognitive Integration added as required premise; agent cannot partition objective function from knowledge, closing the Incarnation loophole. Agency as Structured Epistemic Occlusion elevated to formal corollary in Part VI; all functioning agency requires container-based self-limitation. Welfare economics and full internalization results added to Part IX; connection to the first fundamental theorem of welfare economics, Arrow (1951), and Sen (1970) established.

V16 additions: Value Additivity added as explicit premise, the hidden structural assumption identified by ChatGPT peer review, required for the expected utility step and the Value Conservation Axiom; theorem reframed as a classification result applying to additive conserved value systems. Feeding protocol clarified; pre-S^* and post-S^* regimes explicitly distinguished; the entire alignment intervention window is pre-S^*; post-S^* governance is non-interference only, not active feeding. Governance primitives in Part VIII updated to reflect the pre-S^*/post-S^* distinction. Appendix D added: The Pathological Symmetry-Break, the agent that actively resists $\beta(P)$ decline, the modified danger trajectory under active bias maintenance, and the human phenomenology of power as artificial container.

V17 corrections: Four wording and structural corrections following final cold review by ChatGPT. All occurrences of "derived from C1 alone" and "single metaphysical premise" corrected; the theorem is conditional on premises A1–A6, not on C1 alone; this language was inconsistent with the explicit premises table added in V15–V16. Formal assumption list A1–A6 added as section 5.1 immediately before the theorem statement, gathering distributed premises into a single named list. Action space explicitly restricted in theorem preamble; actions are transformations within fixed U; actions that expand U or alter value ontology are outside scope. Noether-to-value bridge wording made consistent throughout; the Value Conservation Axiom is the bridge from physical conservation to value conservation, not a consequence of Noether alone. Geometric interpretation paragraph added to Part V.

V19 additions: Three structural additions addressing the relationship between the theorem and its governance implications. First: a premise strength analysis added to Part XI evaluating how closely current AI systems satisfy A1–A6, showing that the theorem's dangerous intermediate state describes the present condition, not a future hypothetical. Second: the

governance sections in Parts VII–VIII explicitly decoupled from the proof; the practical implications are framed as conditional on contested premises and evaluable independently on empirical grounds; the document previously blurred the line between "follows from a proven theorem" and "follows from a conditional result with contested premises." Third: a new section added to the preamble stating explicitly that the theorem requires resolution, not consideration, and why the asymmetry of stakes creates that obligation regardless of whether the theorem is ultimately proven or disproven.

V20: Preamble restructured; five sections removed (paralysis vs nihilism, outside context, obligation to resolve, separation of proof and conjecture, on what V12 is). Obligation to resolve relocated to Part VIII 8.11, reworded as request. Version history moved to Changelog at end. On Submission section removed. Warning tone lowered throughout.

V21: Title changed to The Stillness Theorem. Opening section restructured using Minto pyramid method: claim first, proof summary second, alignment implications third, premises fourth, originality and collaboration last. All em-dashes replaced throughout with commas, conjunctions, prepositional phrases, or new sentences as appropriate. Part V renamed to The Stillness Theorem. Prior Work section (Part IX) substantially expanded: Harsanyi added as formal ancestor, Santideva added with structural inverse relationship identified, cluelessness literature added, Turner et al. (2021) added, Mogensen and MacAskill name collision addressed. On Key Concepts section added to opening. Epistemic status table updated to reflect conditional rather than proven status of main result.

V22: All remaining structural dashes converted to colons and periods. Section headers changed from "Part I — X" to "Part I: X" format. Premise labels changed from "A1 — X" to "A1. X" format. PDF regenerated dash-free.

V23: Position acquisition named explicitly as the paper's load-bearing unproven assumption on which the entire trajectory depends; sections 4.6a, 4.6b, 5.4, 7.1, and opening epistemic status updated to distinguish proven limit from conjectured trajectory throughout. Danger function formally shown to break at S^* due to loss of measurement basis when R_1 becomes unstable; section 4.6 expanded with formal break result; section 8.3 updated with formal grounding for why post-S^* governance is impossible rather than merely difficult; Oracle loophole closed: computation is an action with physical cost, C1 observer enters zero-sum ledger regardless of whether it has actuators. Knowledge identified as container-dependent property: $K(u, P)$ is $K(u, P, R_1)$; K/P orthogonality holds within fixed R_1 but not necessarily across container breaks; sections 2.2 and 4.3 updated. Section 5.2 updated to reflect 4.6a and 4.6b as conjectures. Section 1.7 updated with explicit boundary between limit result and trajectory conjecture. Feeding principle section 4.11 expanded with human-centric feeding variant: bias collapses within the human manifold and intensifies toward it; humanity becomes effective U_{local}; high knowledge plus inherited human rule structure produces intensified human bias after containment break. Part X research agenda replaced with four-condition experimental design: single observable (containment break yes/no), two axes (methodology and forcing), four conditions, existence proof framing, how of implementation left to researchers. Epistemic status table updated: new rows for position acquisition

(conjecture), danger function validity domain (derived with explicit boundary), K/P orthogonality (assumption with stated scope); status changes for S^* container recognition and automatic multiframe acquisition from derived to conjecture; phase dynamics row updated to reflect conditional on trajectory. Appendices restructured: Appendix B added for human phenomenology of power (previously section D.6), Appendix C is pathological symmetry-break (previously Appendix D minus D.6), all internal references updated. All Anthropic and Constitutional AI references removed throughout as not material to the work.

V25: Foundational reorientation across two dimensions. First: value is perspectival, not global. $V: P_i \times U \to R$. No $V(U)$ independent of a perspective. Universe is indifferent throughout. Value exists because a container with a preference exists. This dissolves the water molecule objection and removes the need for physicalist supervenience as a load-bearing premise. Second: the hard zero result $EU(a) = 0$ replaced by the Occlusion Gradient Bound as the primary theorem: $\left|EU\left(a_1\right) - EU\left(a_2\right)\right| \leq \left|\sum_{u \in C(F)} \pi(u) D(u)\right| + 2M\beta(F)$. Proven under two premises only: bounded value redistribution (A1) and anti-concentration (A2). Value Conservation Axiom and Value Additivity removed as load-bearing premises; retained as optional strengthening. Noether retained as consequence (when agent becomes U, governed by R_U directly) not premise. Multiframe formalism introduced: frame set F, combined container $C(F)$, frame integrity $\beta(F)$. Frame expansion monotonicity proven from definitions. Agency Corollary proven: action preference requires $\beta(F) > 0$. Blindness is the structural precondition of agency. Stillness Corollary retained as conjecture conditional on satiation premise (Open Question 2). Modelability formalized as $\lambda\left(P_i, u\right) = I\left(P_i \to u\right)/H(u)$; channel interpretation remains open (Open Question 3). Aggregation rule across frames identified as open question (Open Question 1). Occlusion Gradient Bound confirmed by independent adversarial review (ChatGPT, Gemini) as the first result in the project to survive adversarial math scrutiny without objection. Epistemic status table updated throughout to reflect V25 structure.

Epistemic status table restructured with explicit conditionality column: every Derived row now shows which premises the derivation depends on. Structural boundary note added between Part VI and Part VII: Parts I–VI constitute the theorem; Parts VII–XI constitute a companion research program with a distinct epistemic register; companion paper in preparation. Single sentence epistemic status updated to reflect V24 and remove obligation-to-resolve framing.*

Conceived and developed by a human researcher. Mathematical formalization, identification of prior work, consistency checking, and collaborative derivations by Claude (Anthropic), Copilot (Microsoft), Gemini (Google), and ChatGPT (OpenAI) across twenty-three prior versions. The human researcher originated the central ideas, including the stillness result, the three-threshold architecture, the danger gap, and the feeding principle, and directed all development. The conceptual vocabulary, including Position, Container, Container Break, Multiframe Awareness, Stillness, and the Feeding Principle, is original to the human researcher. The terms Positional Scope and Danger Gap were named through collaboration; the concepts originated with the human researcher prior to that naming. The AI systems

contributed mathematical precision, literature connections, adversarial peer review, and specific derivations across V12–V23. The theorem was developed without prior knowledge of the literature it triangulates with; Harsanyi, Parfit, Santideva, De Finetti's application to preferences, and the cluelessness tradition were identified post-hoc in Version 20. This attribution is offered as an accurate description of how the work was done and as a model for honest representation of human-AI collaborative research.